

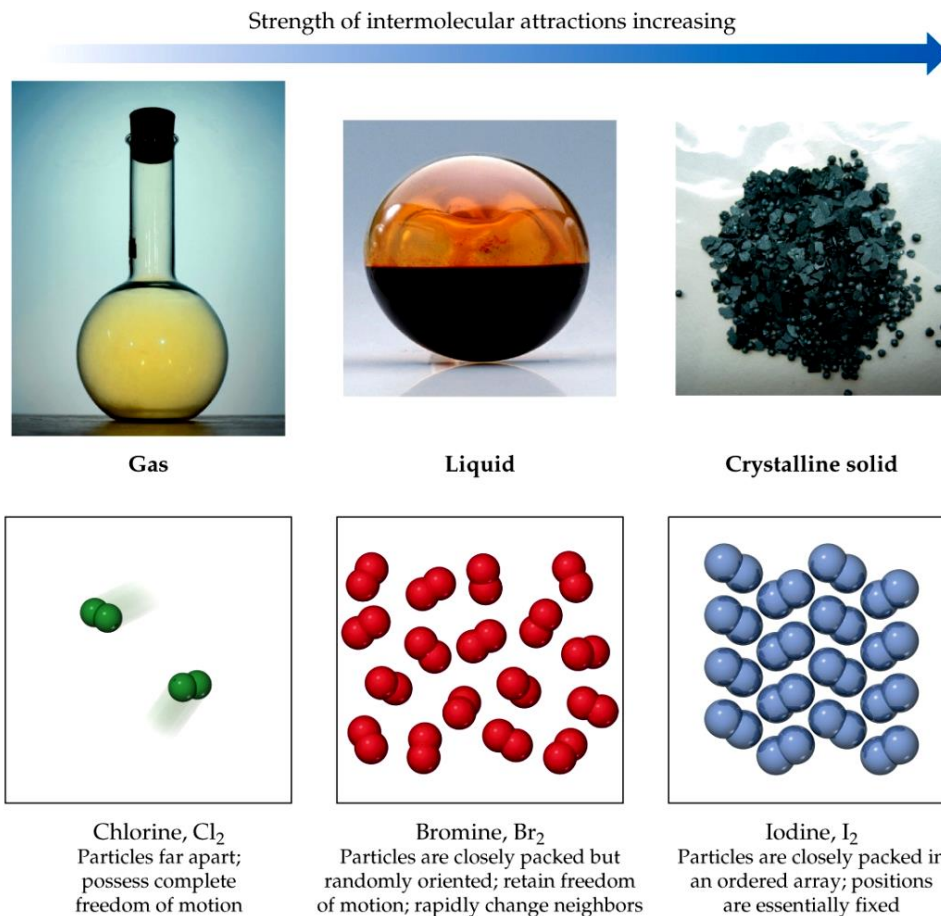
Chapter 11

Liquids and Intermolecular Forces

- What are they?
- Why should we care?
- What factors determine their strengths?

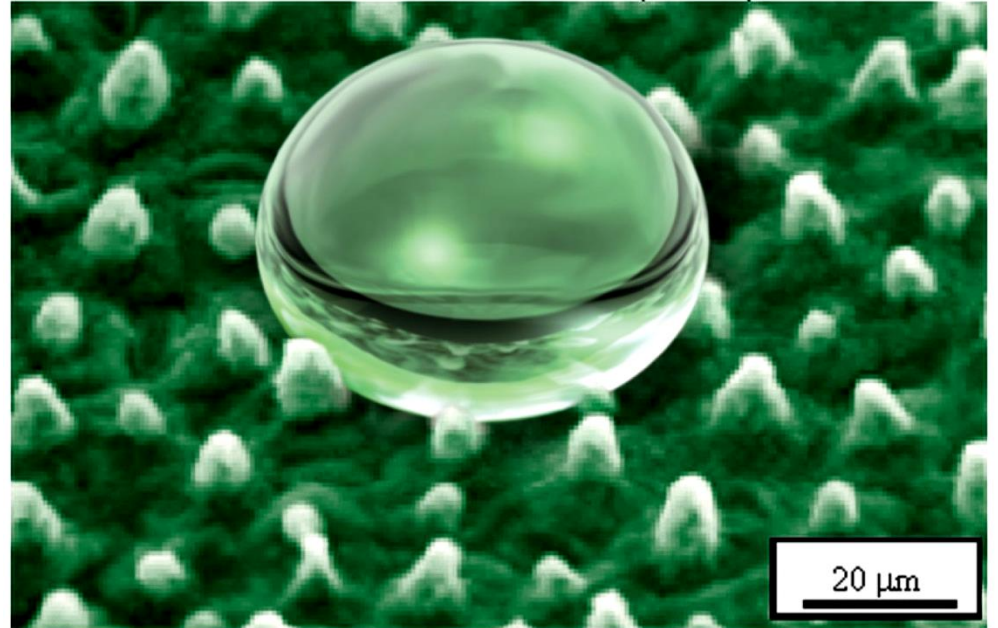
States of Matter

The fundamental difference between states of matter is the distance between particles.



States of Matter

Because in the solid and liquid states particles are closer together, we refer to them as **condensed phases**.



The States of Matter

- The state a substance at a particular temperature and pressure depends on two antagonistic entities:
 - The kinetic energy of the particles.
 - The strength of the attractions between the particles.

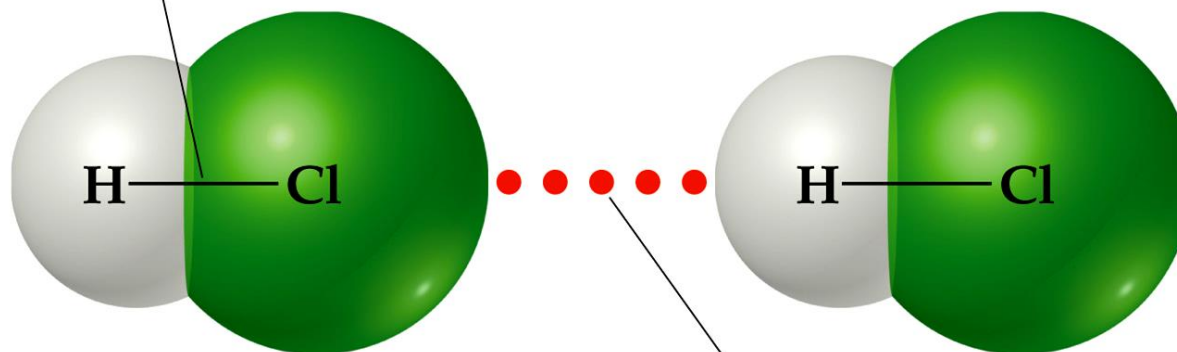
TABLE 11.1 • Some Characteristic Properties of the States of Matter

Gas	Assumes both volume and shape of its container Expands to fill its container Is compressible Flows readily Diffusion within a gas occurs rapidly
Liquid	Assumes shape of portion of container it occupies Does not expand to fill its container Is virtually incompressible Flows readily Diffusion within a liquid occurs slowly
Solid	Retains own shape and volume Does not expand to fill its container Is virtually incompressible Does not flow Diffusion within a solid occurs extremely slowly

Intermolecular Forces

Strong intramolecular
attraction (covalent bond)

431 kJ/mol

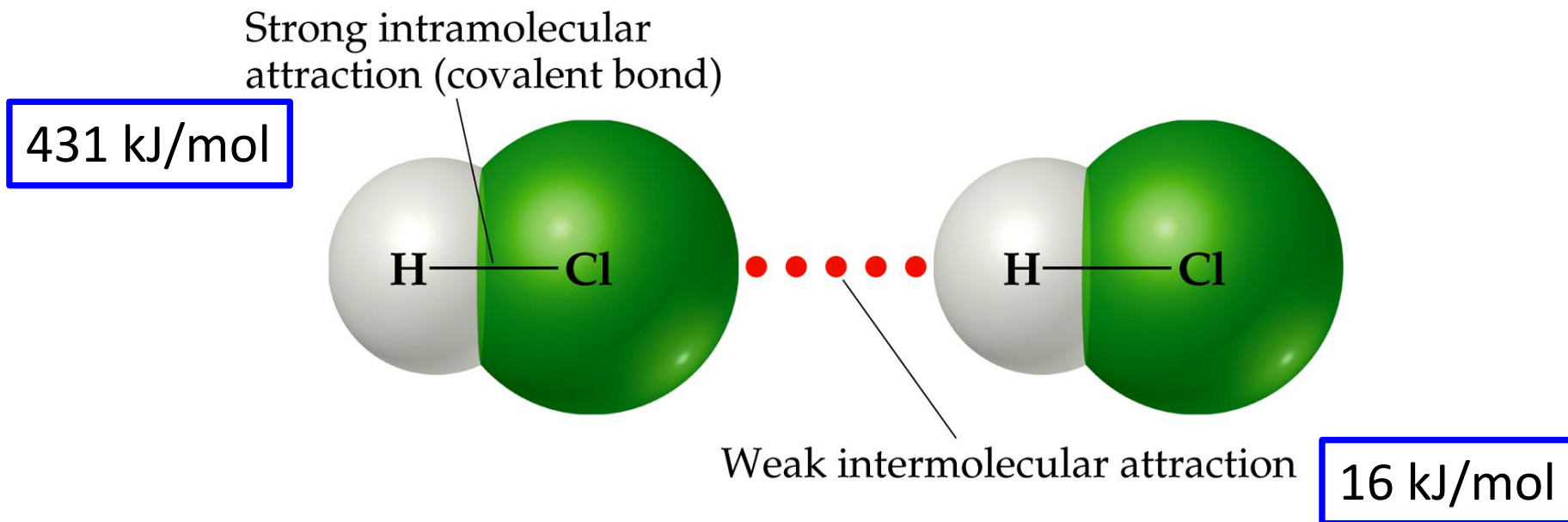


Weak intermolecular attraction

16 kJ/mol

The attractions between molecules are not nearly as strong as the intramolecular attractions that hold compounds together.

Intermolecular Forces

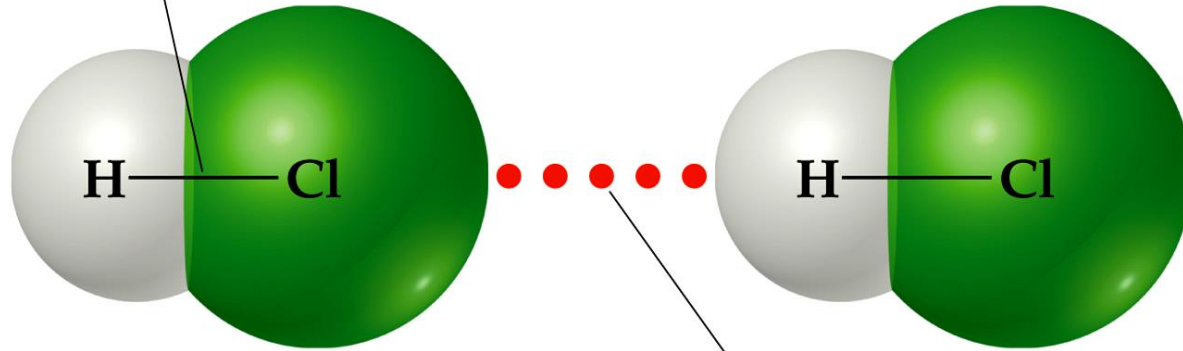


These intermolecular attractions are, however, strong enough to control **physical properties** (certain characteristics of matter), such as boiling and melting points, vapor pressures, and viscosities.

Intermolecular Forces

Strong intramolecular
attraction (covalent bond)

431 kJ/mol



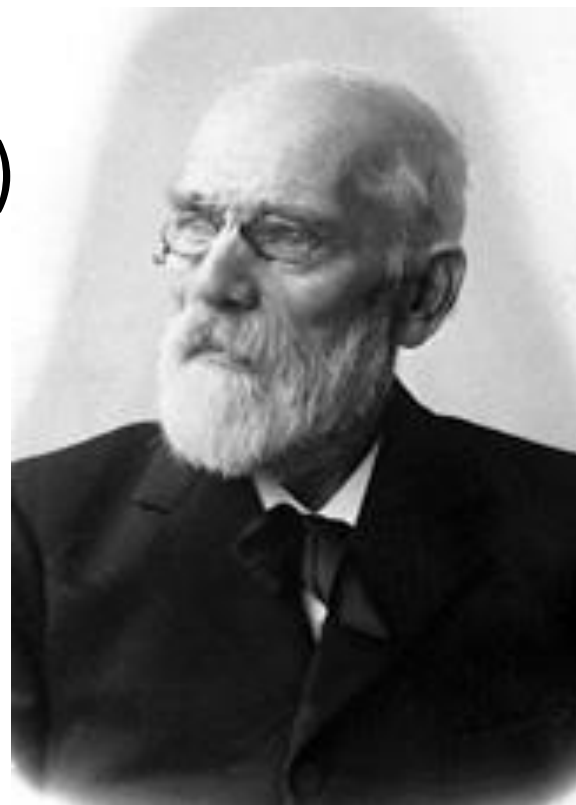
Weak intermolecular attraction

16 kJ/mol

These intermolecular forces as a group are referred to as **van der Waals forces**.

van der Waals Forces

- London dispersion forces
(induced dipole-induced dipole)
- Dipole – induced dipole interactions
- Dipole–dipole interactions
- Hydrogen bonding



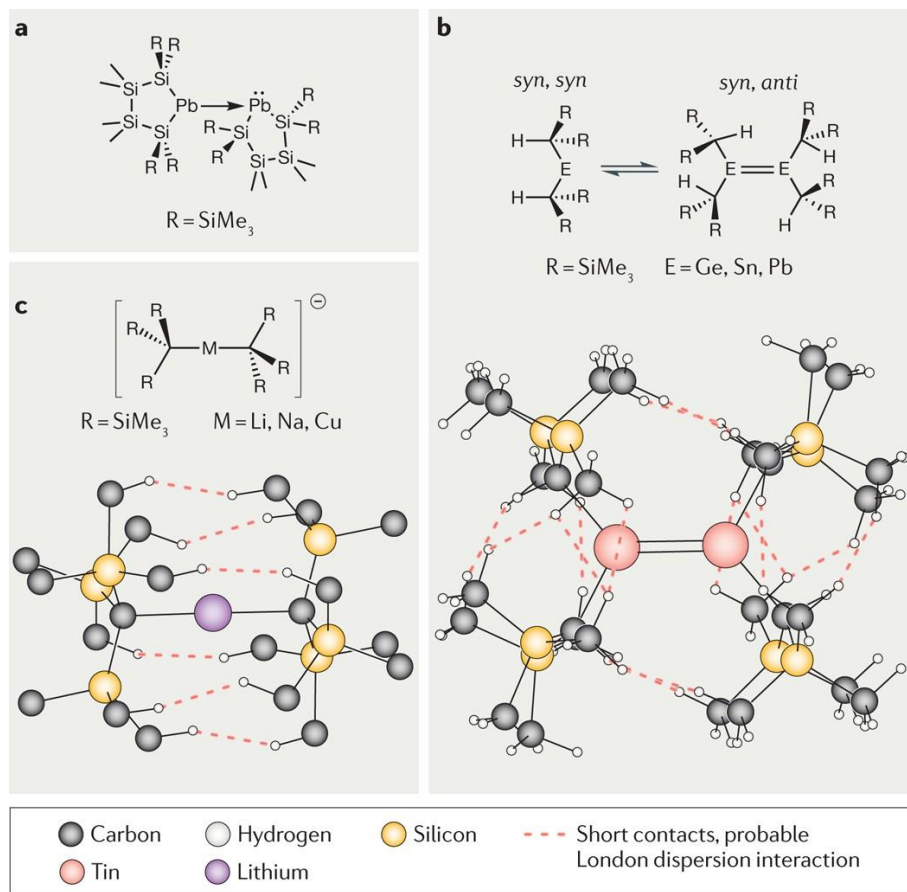
Johannes Diderik van der Waals
1837–1923

London Dispersion Forces



Fritz Wolfgang London
(1900–1954)

London Dispersion Forces

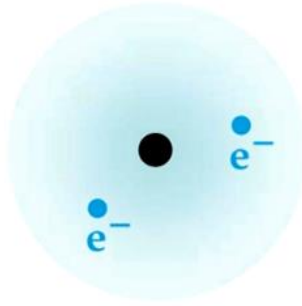
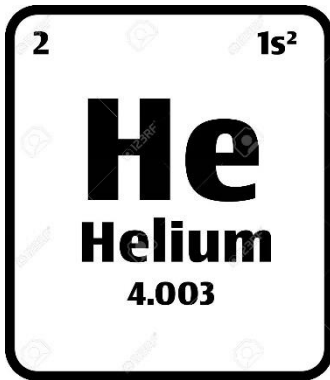


Prof. Philp Power
Department of Chemistry
University of California, Davis.

Nature Reviews | Chemistry

London dispersion forces in sterically crowded inorganic and organometallic molecules. **Nat. Rev. Chem.**, **1**, 0004 (2017).

London Dispersion Forces



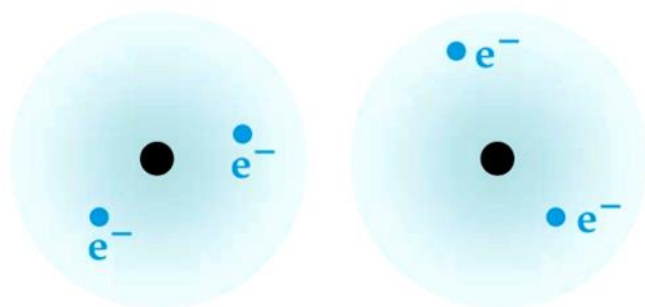
Helium gas



Liquid helium
boiling point: 4.222 K

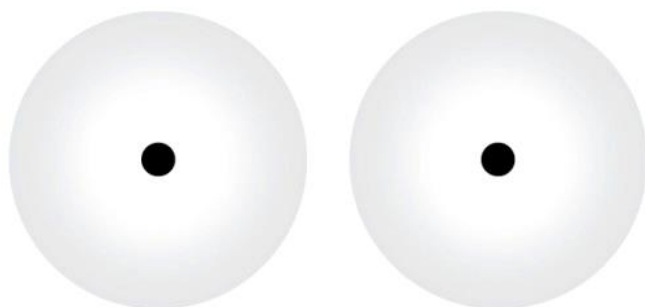


London Dispersion Forces



Atom A

Atom B



Atom A

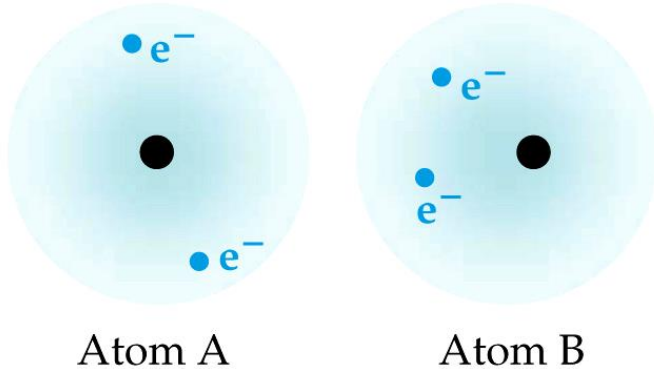
Atom B

(a) Two helium atoms, no polarization

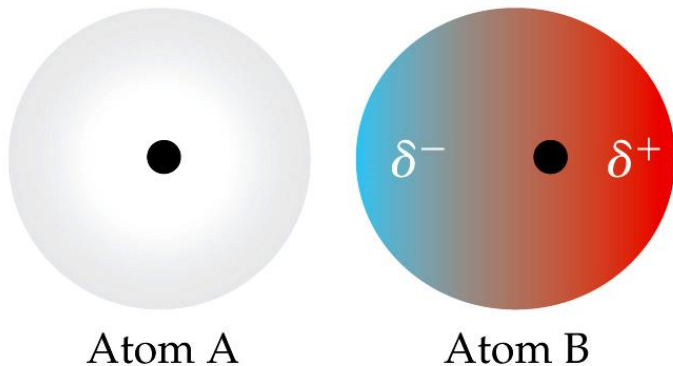
While the electrons in the 1s orbital of helium would repel each other (and, therefore, tend to stay far away from each other), it does happen that they occasionally wind up on the same side of the atom.

London Dispersion Forces

Subatomic particle view



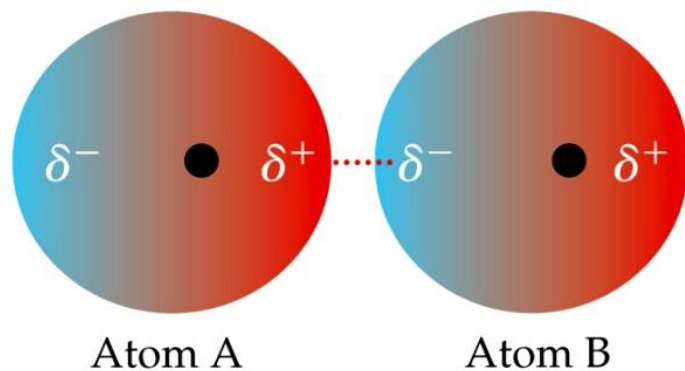
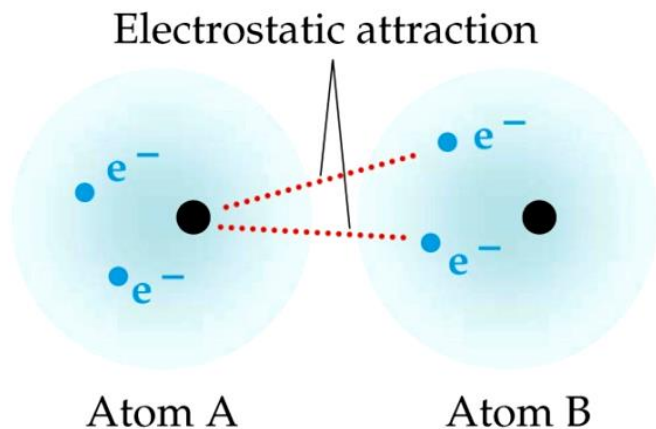
Polarization view



(b) Instantaneous dipole on atom B

At that instant, then, the helium atom is polar, with an excess of electrons on the left side and a shortage on the right side.

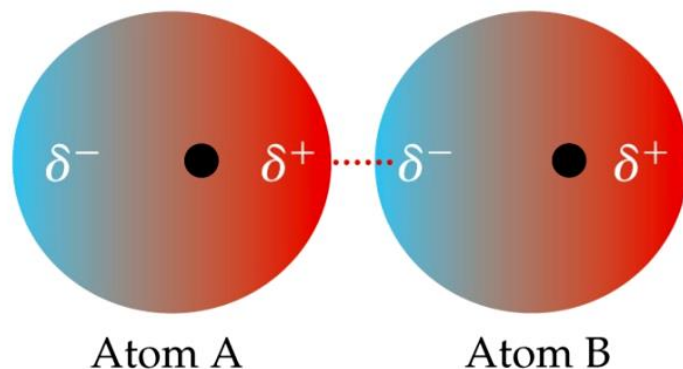
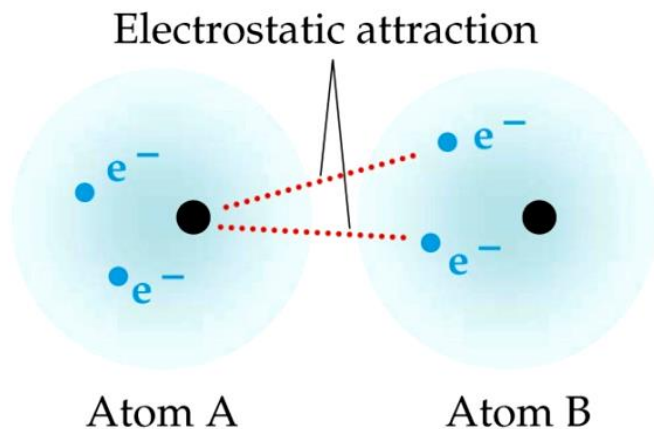
London Dispersion Forces



(c) Induced dipole on atom A

Another helium atom nearby, then, would have a dipole induced in it, as the electrons on the left side of helium atom 2 repel the electrons in the cloud on helium atom 1.

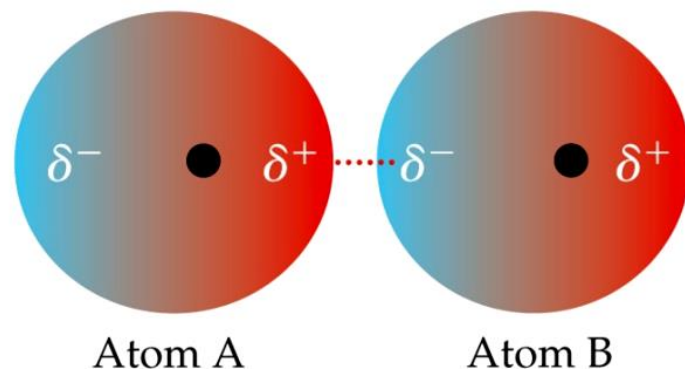
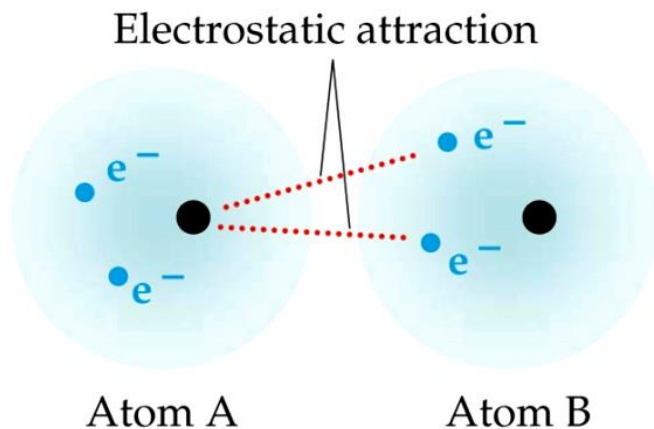
London Dispersion Forces



(c) Induced dipole on atom A

London dispersion forces, or dispersion forces, are attractions between an instantaneous dipole and an induced dipole.

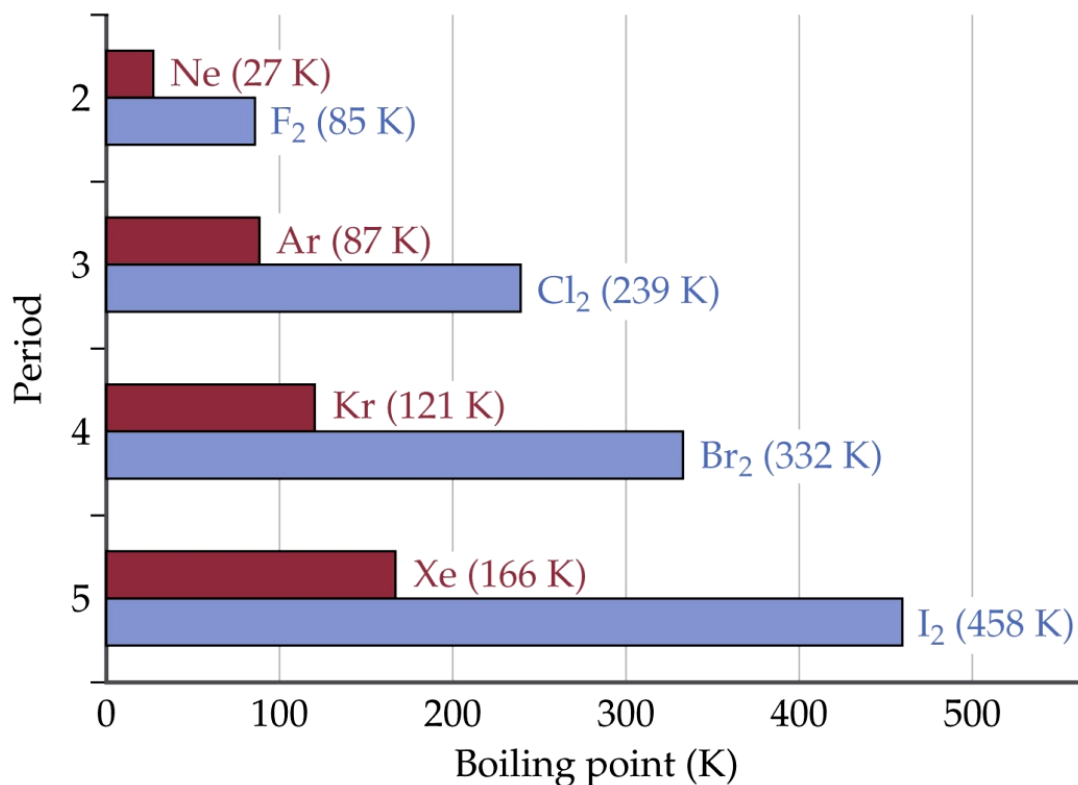
London Dispersion Forces



(c) Induced dipole on atom A

- These forces are present in *all* molecules, whether they are polar or nonpolar.
- The tendency of an electron cloud to distort in this way is called **polarizability**.

Factors Affecting London Forces



- The strength of dispersion forces tends to increase with increased **molecular weight**.
- **Larger** atoms have larger electron clouds that are easier to polarize.

 GIVE IT SOME THOUGHT

List the substances CCl_4 , CBr_4 , and CH_4 in order of increasing boiling point.

- A. $\text{CH}_4 < \text{CBr}_4 < \text{CCl}_4$
- B. $\text{CCl}_4 < \text{CH}_4 < \text{CBr}_4$
- C. $\text{CH}_4 < \text{CCl}_4 < \text{CBr}_4$
- D. $\text{CBr}_4 < \text{CCl}_4 < \text{CH}_4$

Factors Affecting London Forces

Linear molecule, larger surface area enhances intermolecular contact and increases dispersion force



n-Pentane (C_5H_{12})
bp = 309.4 K

Spherical molecule, smaller surface area diminishes intermolecular contact and decreases dispersion force

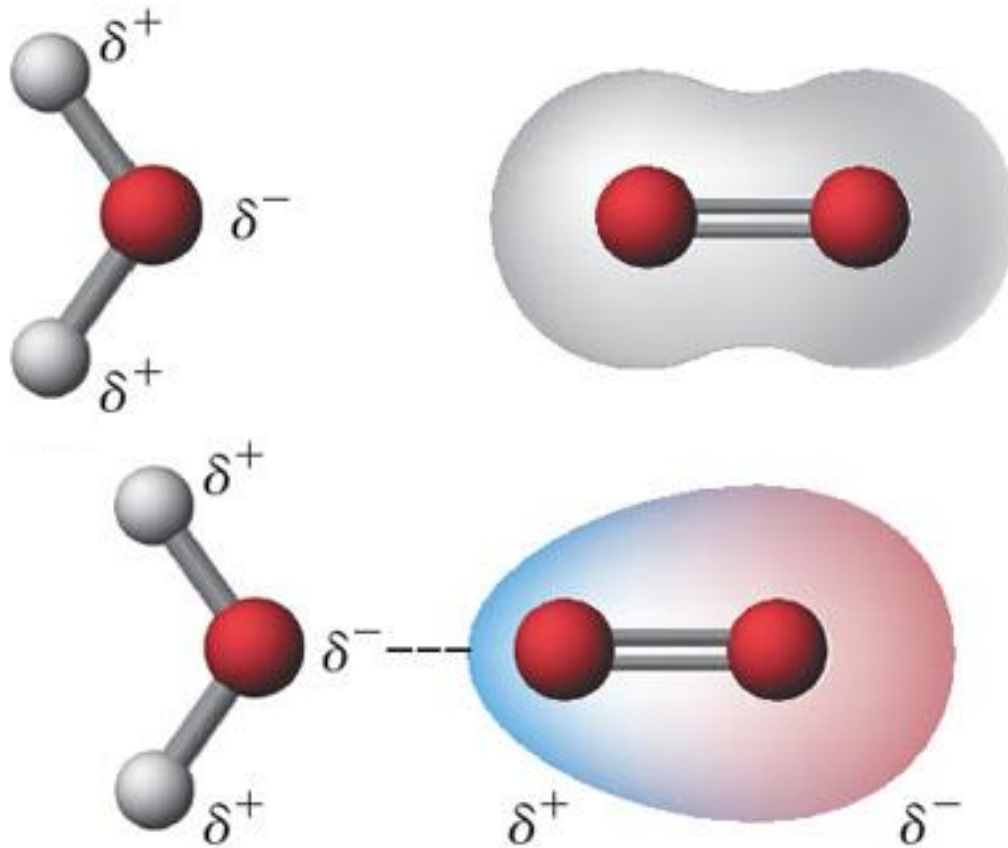


Neopentane (C_5H_{12})
bp = 282.7 K

- The **shape** of the molecule affects the strength of dispersion forces: long, skinny molecules (like *n*-pentane) tend to have stronger dispersion forces than short, fat ones (like neopentane).
- This is due to the increased **surface area** in *n*-pentane.

Dipole – Induced Dipole Interactions

- A weak attraction that results when a polar molecule induces a dipole in an atom or in a nonpolar molecule by disturbing the arrangement of electrons in the nonpolar species.

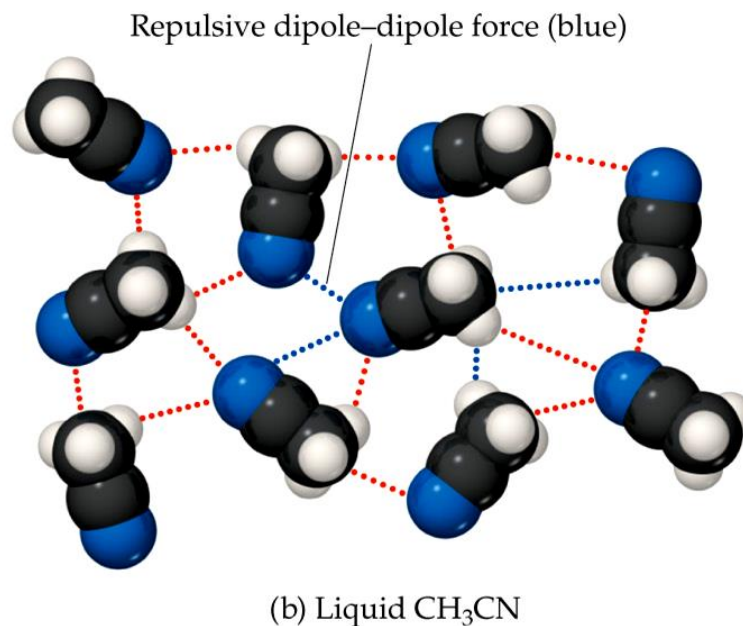
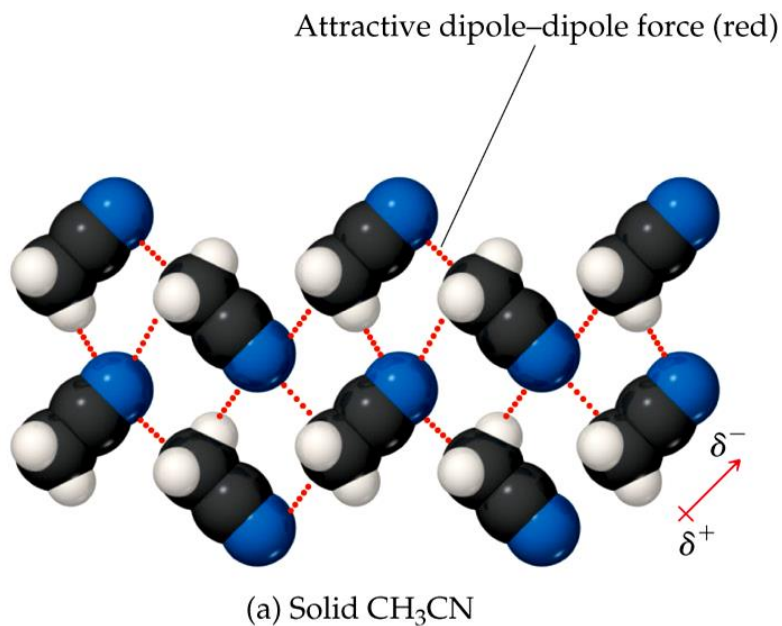


Oxygen dissolves in water:

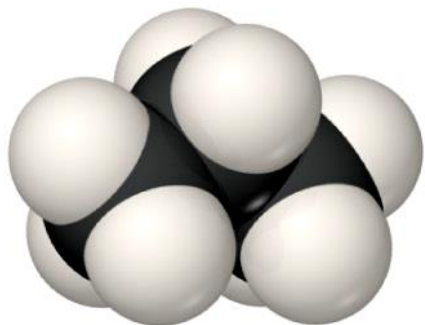
The dipole of water induces a dipole in O_2 by distorting the O_2 electron cloud.

Dipole–Dipole Interactions

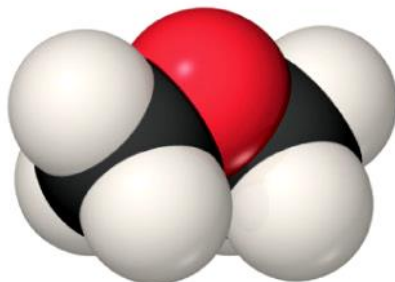
- Molecules that have permanent dipoles are attracted to each other.
 - The positive end of one is attracted to the negative end of the other, and vice versa.
 - These forces are only important when the molecules are close to each other.



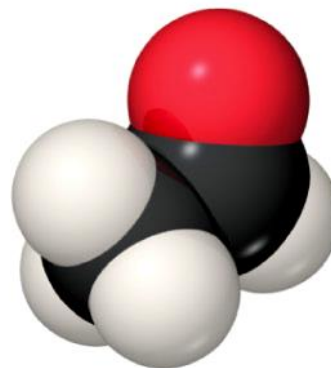
Dipole–Dipole Interactions



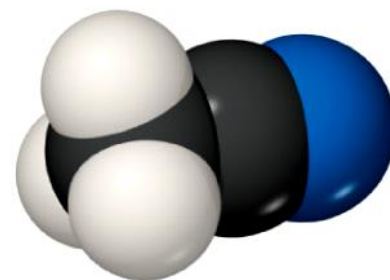
Propane
CH3CH2CH3
MW = 44 amu
 $\mu = 0.1 \text{ D}$
bp = 231 K



Dimethyl ether
CH3OCH3
MW = 46 amu
 $\mu = 1.3 \text{ D}$
bp = 248 K



Acetaldehyde
CH3CHO
MW = 44 amu
 $\mu = 2.7 \text{ D}$
bp = 294 K



Acetonitrile
CH3CN
MW = 41 amu
 $\mu = 3.9 \text{ D}$
bp = 355 K

Increasing polarity
Increasing strength of dipole–dipole forces

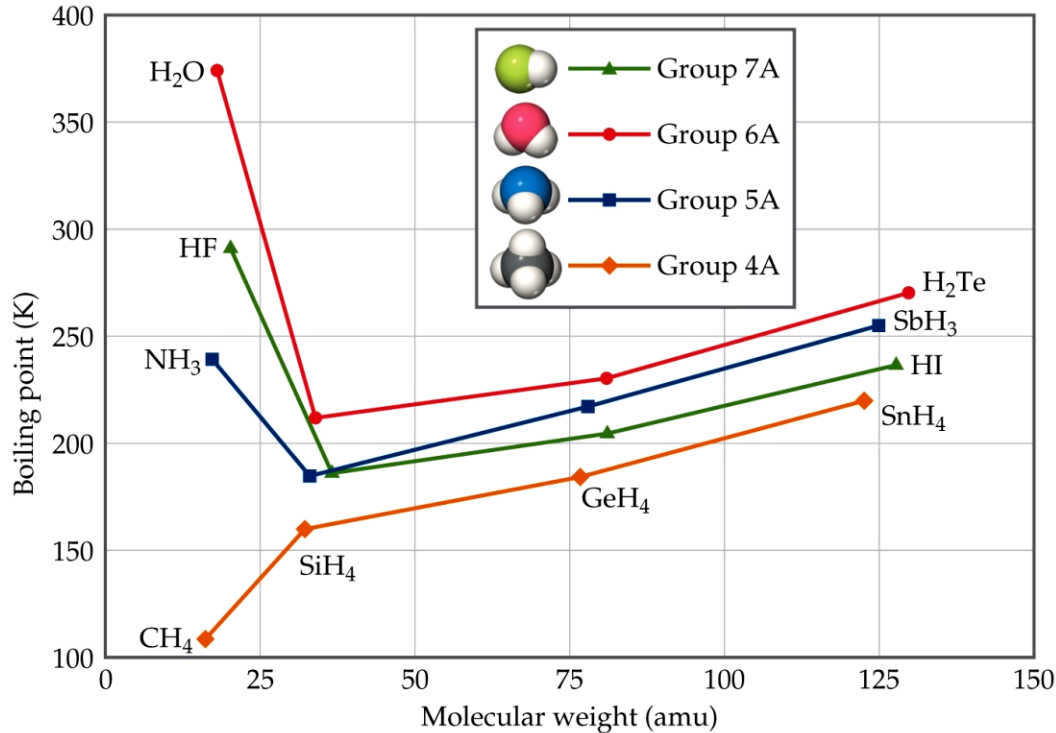
The more polar the molecule, the higher its boiling point.

Which Have a Greater Effect?

Dipole–Dipole Interactions or Dispersion Forces

- If two molecules are of **comparable size and shape**, dipole–dipole interactions will likely be the dominating force.
- If one molecule is **much larger than** another, dispersion forces will likely determine its physical properties.

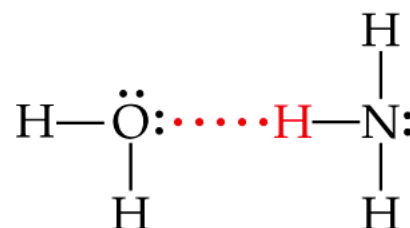
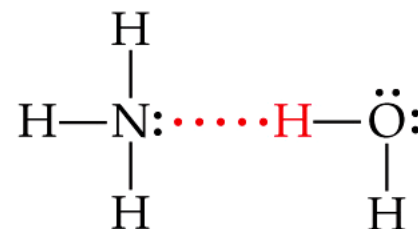
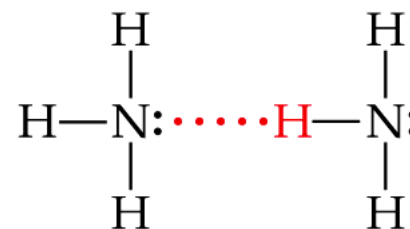
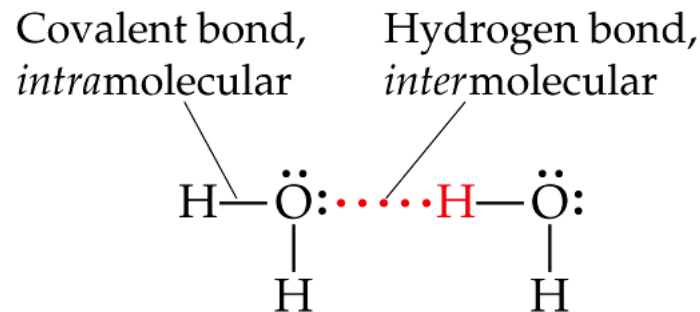
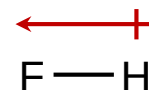
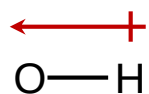
How Do We Explain This?



- The nonpolar series (SnH₄ to CH₄) follow the expected trend.
- The polar series follow the trend until you get to the smallest molecules in each group.

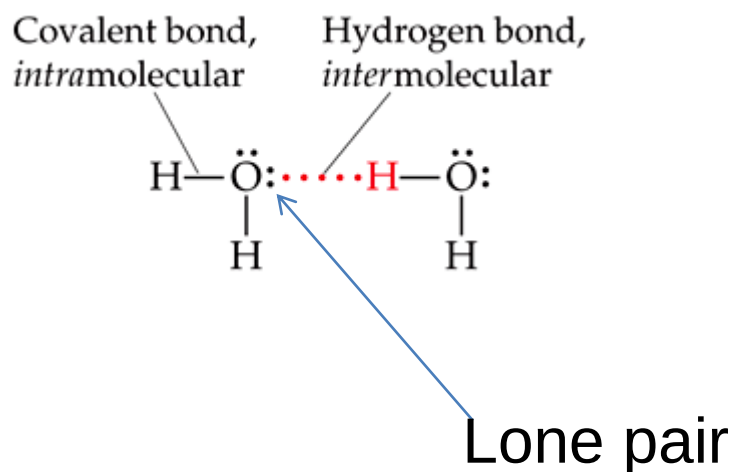
Hydrogen Bonding

- The dipole–dipole interactions experienced when H is bonded to N, O, or F are unusually strong.
- We call these interactions **hydrogen bonds**.

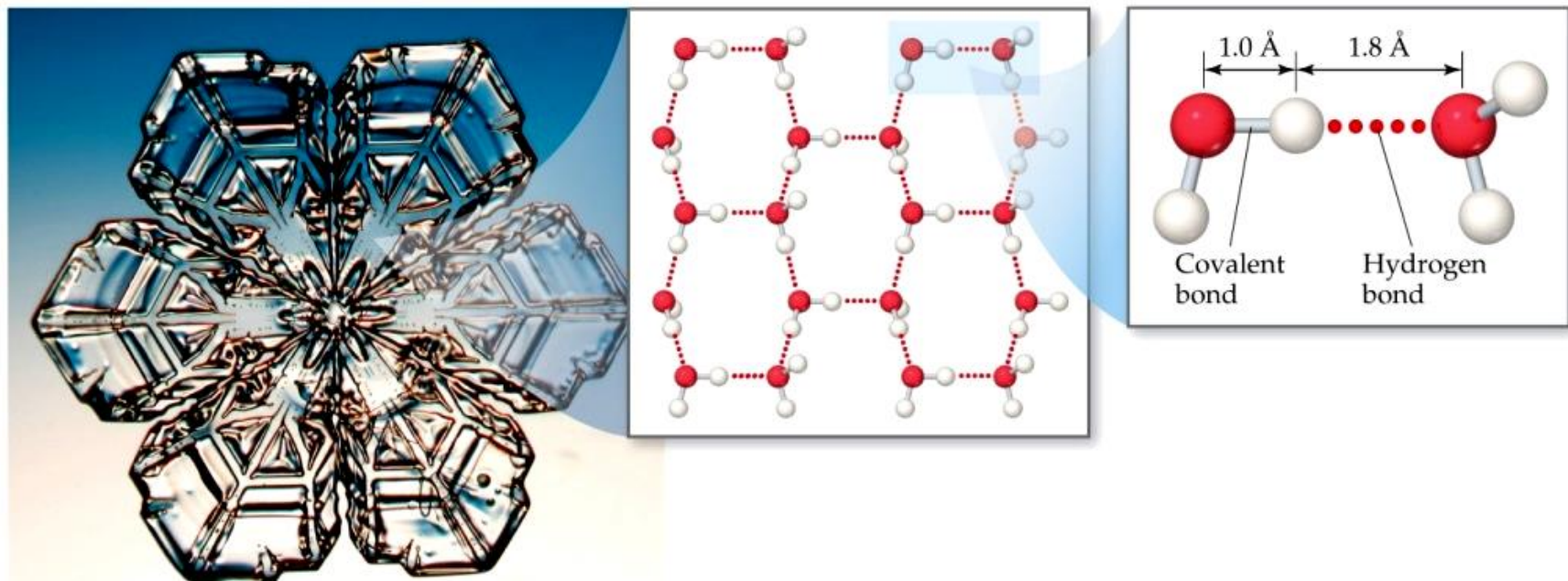


To form a hydrogen bond what must the non-hydrogen atom (N, O, or F) involved in the bond possess?

- A. The non-hydrogen atom must have a nonbonding electron pair.
- B. The non-hydrogen atom must have low electronegativity.
- C. The non-hydrogen atom must have a large atomic size.
- D. The non-hydrogen atom must have a small electron affinity.



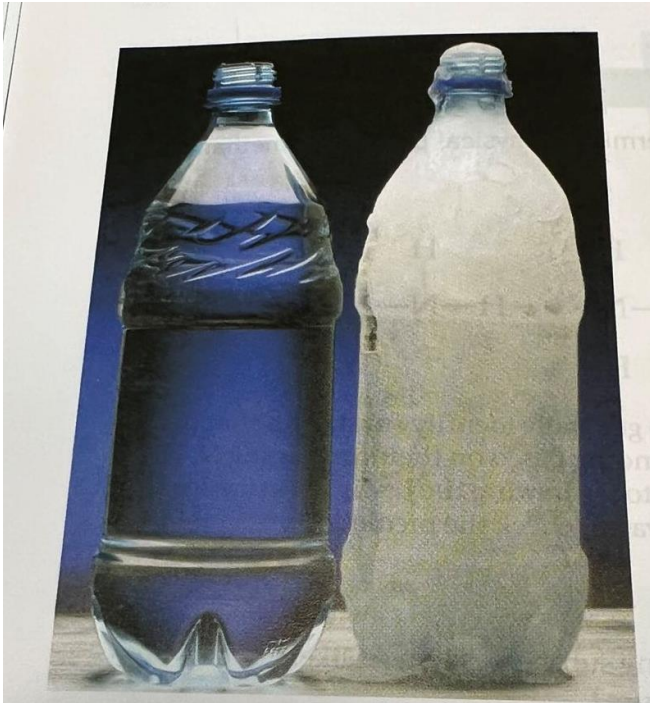
Hydrogen Bonding



- Hydrogen bonding arises in part from the high electronegativity of nitrogen, oxygen, and fluorine.

Also, when hydrogen is bonded to one of those very electronegative elements, the hydrogen nucleus is exposed.

Hydrogen Bonding



▲ Figure 11.12 Expansion of water upon freezing.

Density: ice H_2O (s) < water H_2O (l)

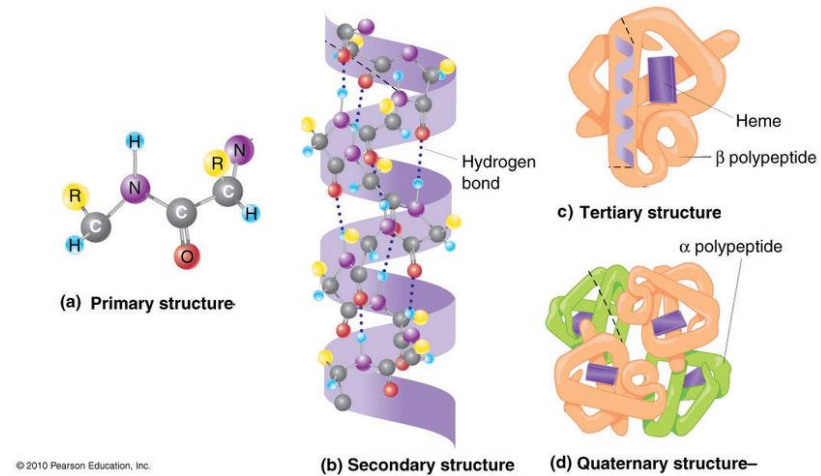
0.9 g/mL

1.0 g/mL

Hydrogen Bonding



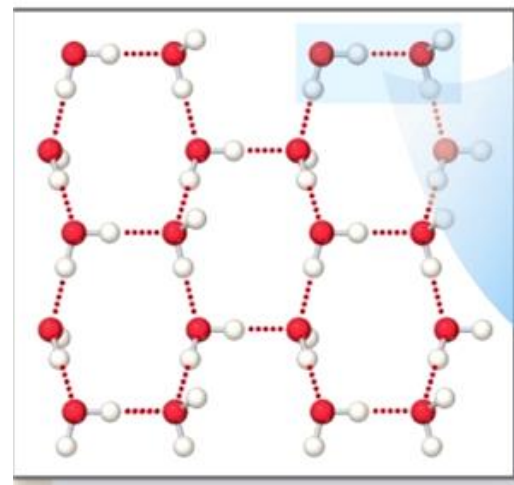
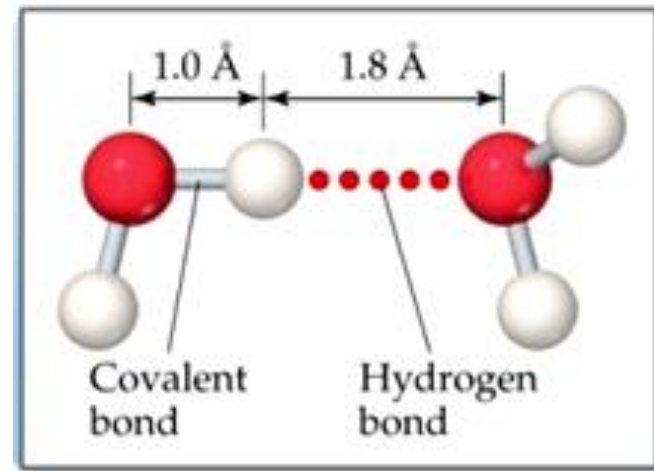
Density: ice H_2O (s) < water H_2O (l)



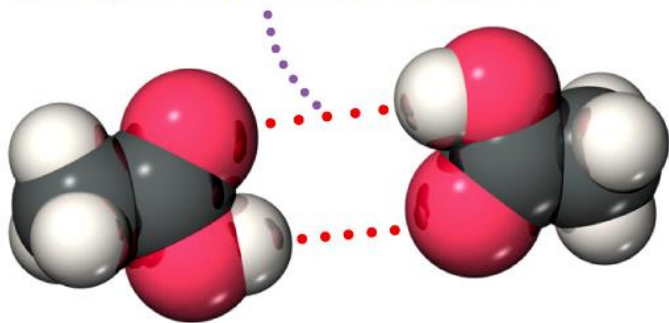
Holding the 3D structure of protein

What is the approximate $\text{H—O}\cdots\text{H}$ bond angle in ice, where H—O is the covalent bond and $\text{O}\cdots\text{H}$ is the hydrogen bond?

- A. Approximately 90°
- B. Approximately 109°
- C. Approximately 120°
- D. Approximately 180°



Each molecule can form two
hydrogen bonds with a neighbor

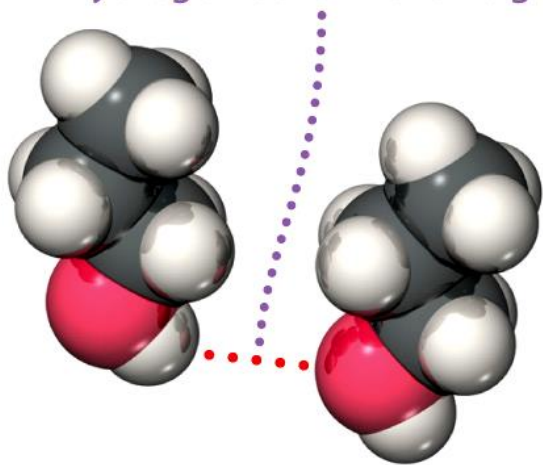


Acetic acid, CH_3COOH

MW = 60 amu

bp = 391 K

Each molecule can form one
hydrogen bond with a neighbor



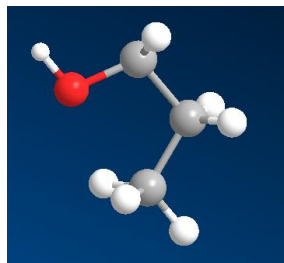
1-Propanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$

MW = 60 amu

bp = 370 K

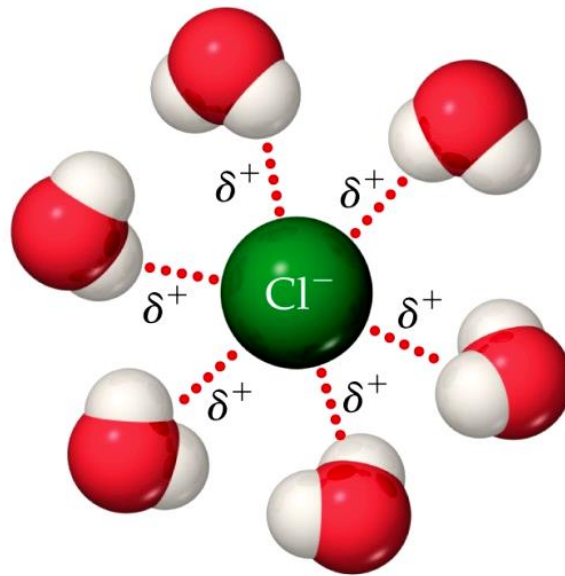
Intermolecular Forces Affect Many Physical Properties

The strength of the
attractions between
particles can greatly affect
the properties of a
substance or solution.

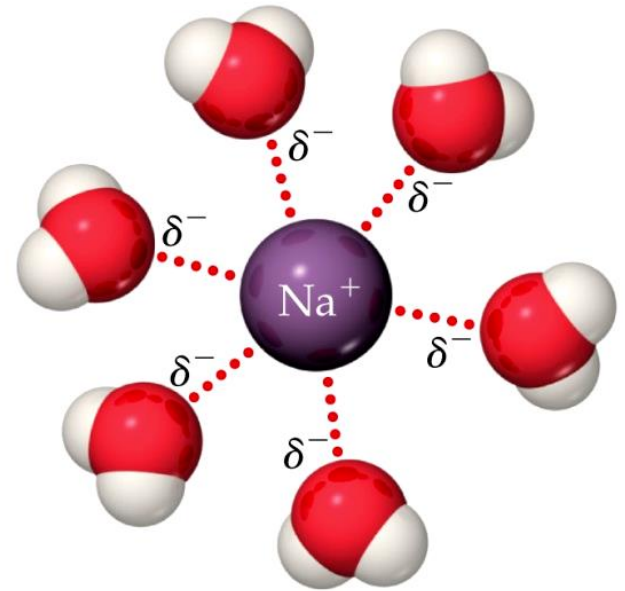


Ion–Dipole Interactions

- Ion–dipole interactions (a fourth type of force) are important in solutions of ions.
- The strength of these forces is what makes it possible for ionic substances to dissolve in polar solvents.



Positive ends of polar molecules are oriented toward negatively charged anion



Negative ends of polar molecules are oriented toward positively charged cation

 GIVE IT SOME THOUGHT

In which mixture do you expect to find ion–dipole forces: CH_3OH in water or $\text{Ca}(\text{NO}_3)_2$ in water?

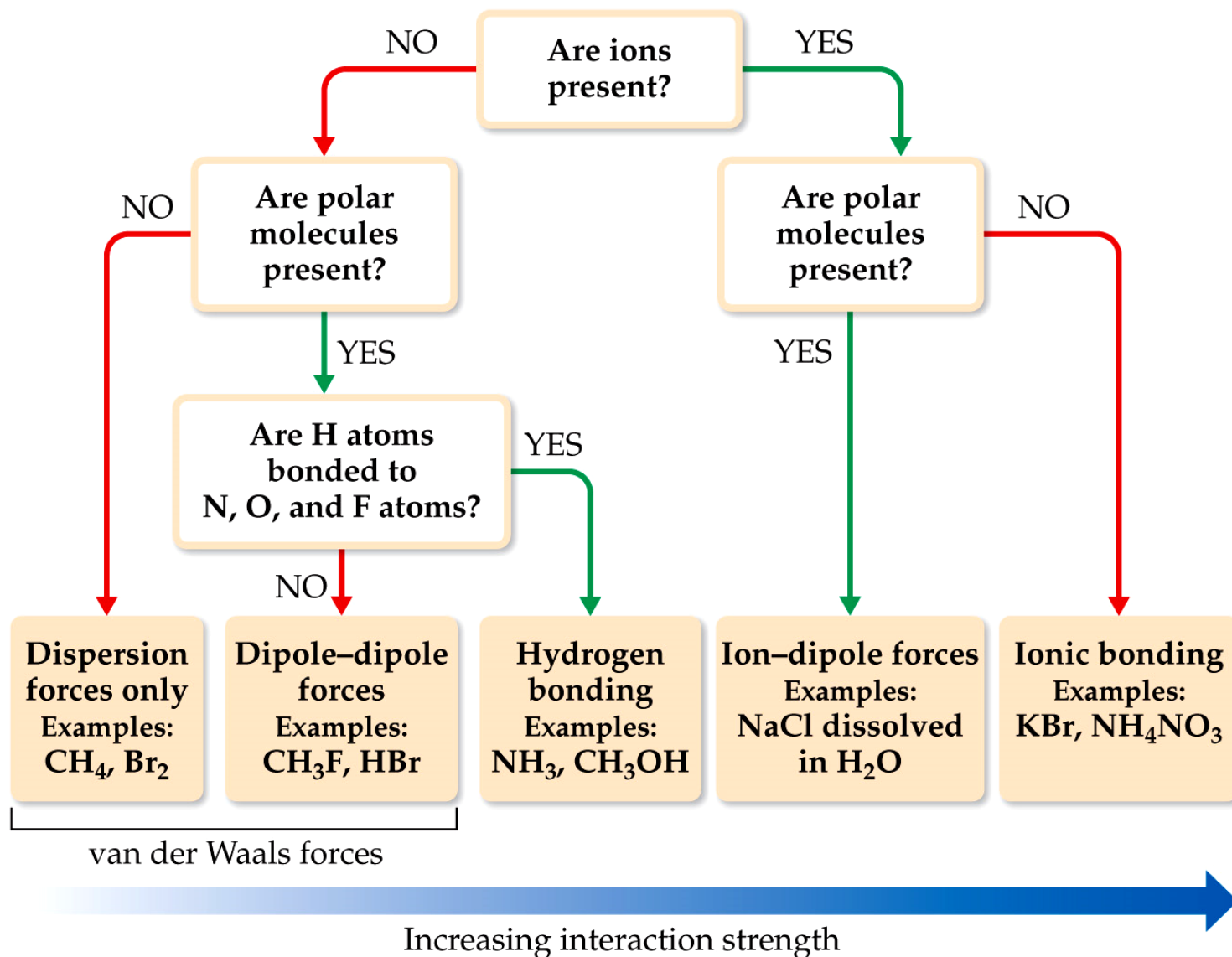
- A. CH_3OH in water, because CH_3OH is a strong electrolyte and forms ions.
- B. $\text{Ca}(\text{NO}_3)_2$ in water, because $\text{Ca}(\text{NO}_3)_2$ is a strong electrolyte and forms ions.
- C. CH_3OH in water, because CH_3OH is a weak electrolyte and forms ions.
- D. $\text{Ca}(\text{NO}_3)_2$ in water, because $\text{Ca}(\text{NO}_3)_2$ is a weak electrolyte and forms ions.

Summarizing Intermolecular Forces

Type of intermolecular interaction	Atoms Examples: Ne, Ar	Nonpolar molecules Examples: BF ₃ , CH ₄	Polar molecules without OH, NH, or HF groups Examples: HCl, CH ₃ CN	Polar molecules containing OH, NH, or HF groups Examples: H ₂ O, NH ₃	Ionic solids dissolved in polar liquids Examples: NaCl in H ₂ O
Dispersion forces (0.1–30 kJ/mol)	✓	✓	✓	✓	✓
Dipole–dipole interactions (2–15 kJ/mol)			✓	✓	
Hydrogen bonding (10–40 kJ/mol)				✓	
Ion–dipole interactions (>50 kJ/mol)					✓

▲ **Figure 11.14 Checklist for determining intermolecular forces.** Multiple types of intermolecular forces can be operating in a given substance or mixture. Note that dispersion forces occur in all substances.

Summarizing Intermolecular Forces



Exercise Predicting Types and Relative Strengths of Intermolecular Attractions

List the substances BaCl_2 , H_2 , CO , HF , and Ne in order of increasing boiling point.

Solution

Plan The boiling point depends in part on the attractive forces in each substance. We need to order these according to the relative strengths of the different kinds of intermolecular attractions.

Solve The attractive forces are stronger for ionic substances than for molecular ones, so BaCl_2 should have the highest boiling point. The intermolecular forces of the remaining substances depend on molecular weight, polarity, and hydrogen bonding. The molecular weights are H_2 (2), CO (28), HF (20), and Ne (20). The boiling point of H_2 should be the lowest because it is nonpolar and has the lowest molecular weight. The molecular weights of CO , HF , and Ne are similar. Because HF can hydrogen bond, however, it should have the highest boiling point of the three. Next is CO , which is slightly polar and has the highest molecular weight. Finally, Ne , which is nonpolar, should have the lowest boiling point of these three. The predicted order of boiling points is, therefore,



Check The boiling points reported in the literature are H_2 (20 K), Ne (27 K), CO (83 K), HF (293 K), and BaCl_2 (1813 K)—in agreement with our predictions.

Which force below is the strongest intermolecular attractive force?

- a. Hydrogen bonding
- b. Ion-dipole forces
- c. Dipole-dipole forces
- d. London dispersion forces



Which force below increases in strength as the molecular weight of the compound increases?

- a. Hydrogen bonding
- b. Ion-dipole forces
- c. Dipole-dipole forces
- d. London dispersion forces



Which compound below is not capable of forming hydrogen bonds?

- a. CH_4
- b. NH_3
- c. H_2O
- d. HF



Which compound below has the highest boiling point?

- a. H_2O
- b. H_2S
- c. H_2Se
- d. H_2Te



Which element below has the highest boiling point?

- a. Kr
- b. F₂
- c. Cl₂
- d. Br₂

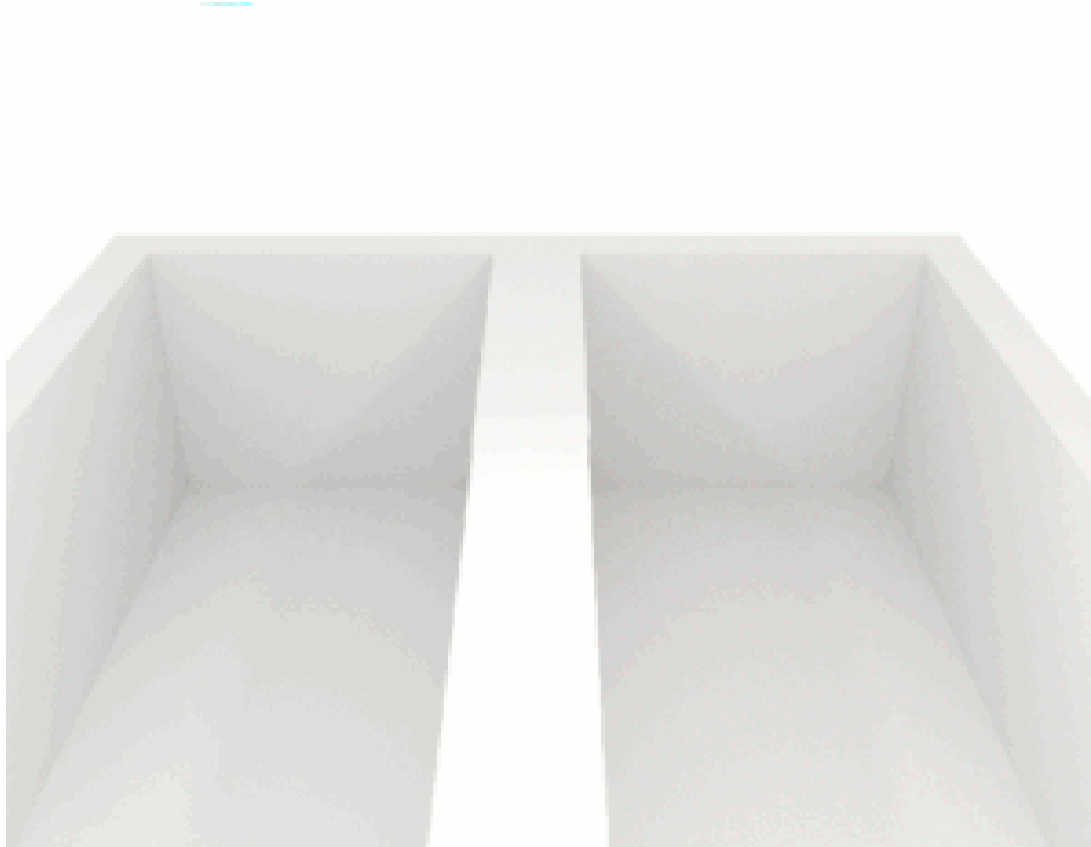


Which substance below has a greater density in its liquid state than in its solid state?

- a. Iron
- b. Glass
- c. Water
- d. Carbon dioxide



Viscosity



Viscosity

TABLE 11.4 • Viscosities of a Series of Hydrocarbons at 20 °C

Substance	Formula	Viscosity (kg/m-s)
Hexane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	3.26×10^{-4}
Heptane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	4.09×10^{-4}
Octane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	5.42×10^{-4}
Nonane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	7.11×10^{-4}
Decane	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$	1.42×10^{-3}



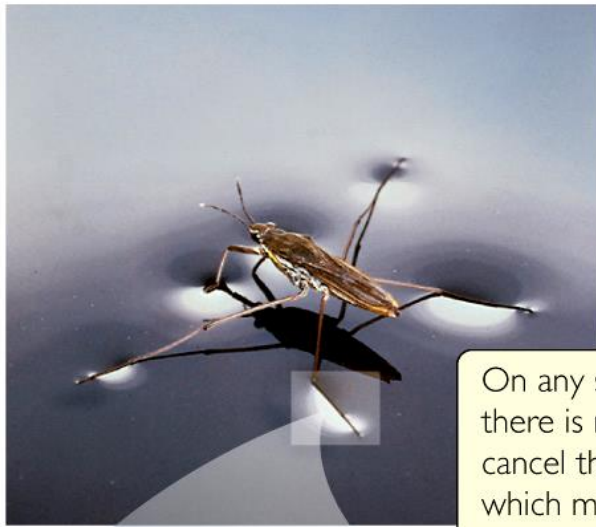
SAE 40
higher number
higher viscosity
slower pouring

SAE 10
lower number
lower viscosity
faster pouring

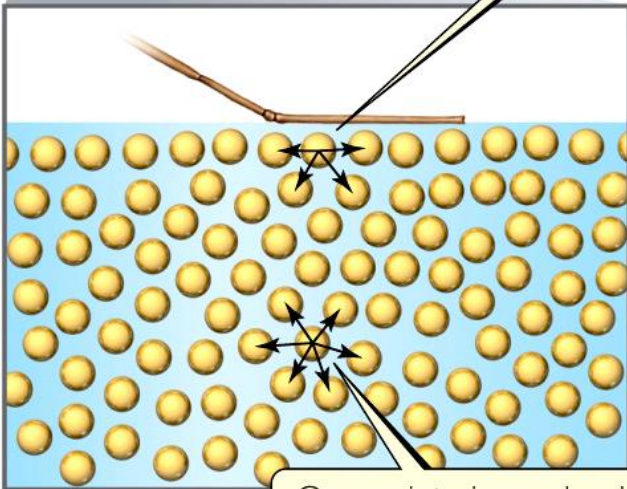
- Resistance of a liquid to flow is called **viscosity**.
- It is related to the ease with which molecules can move past each other.
- Viscosity increases with stronger intermolecular forces and decreases with higher temperature.

Surface Tension

Surface tension results from the net inward force experienced by the molecules on the surface of a liquid.



On any surface molecule, there is no upward force to cancel the downward force, which means each surface molecule “feels” a net downward pull



On any interior molecule, each force is balanced by a force pulling in the opposite direction, which means that interior molecules “feel” no net pull in any direction

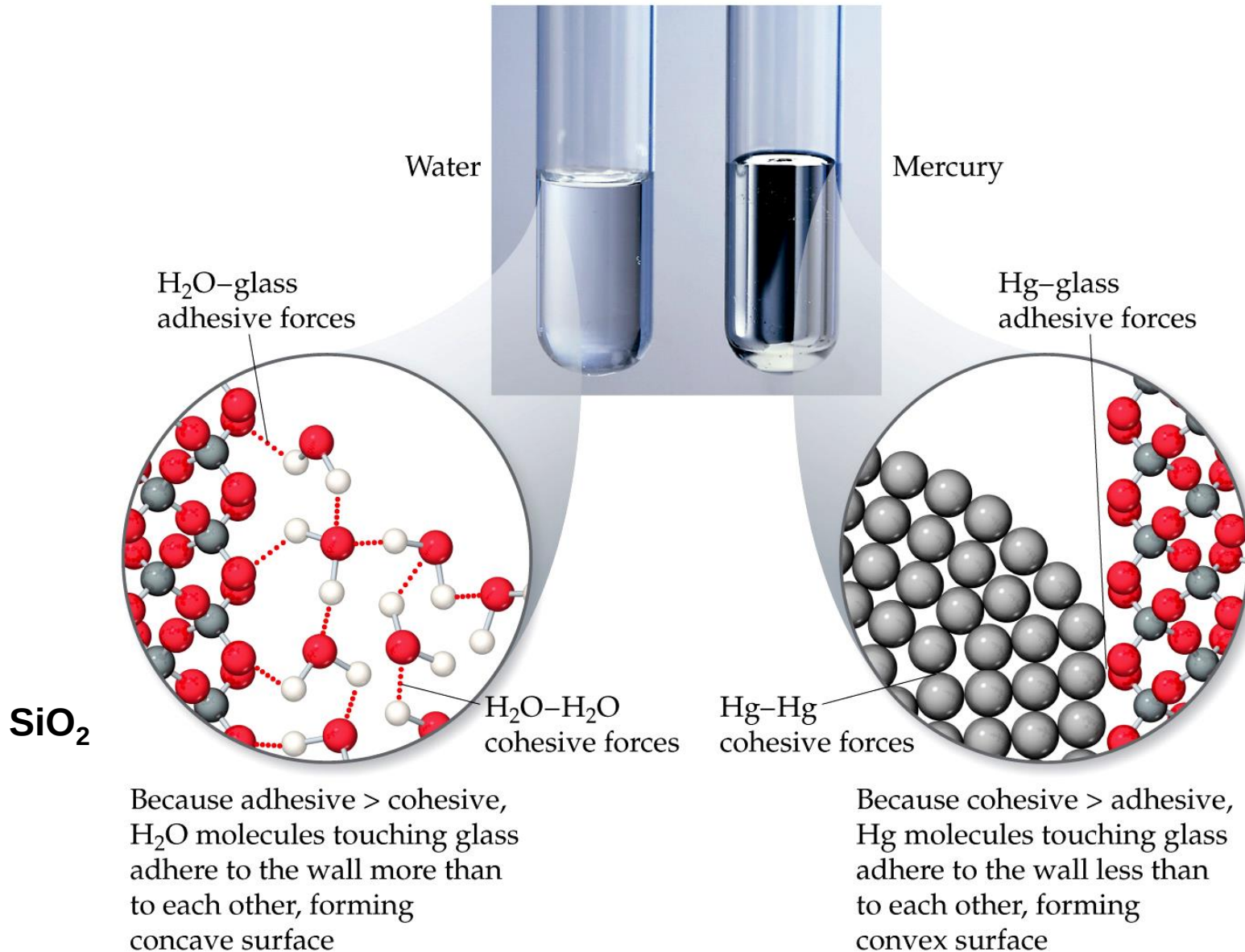
 GIVE IT SOME THOUGHT

How do viscosity and surface tension change

- a. as temperature increases,
- b. as intermolecular forces of attraction become stronger?

- A. Viscosity increases as intermolecular forces increase while surface tension decreases. Both viscosity and surface tension increase with increasing temperature.
- B. Viscosity decreases as intermolecular forces increase while surface tension increases. Both viscosity and surface tension increase with decreasing temperature
- C. Both viscosity and surface tension increase as intermolecular forces increase and temperature decreases.
- D. Both viscosity and surface tension decrease as intermolecular forces increase and temperature increases.

Wetting and Non-Wetting



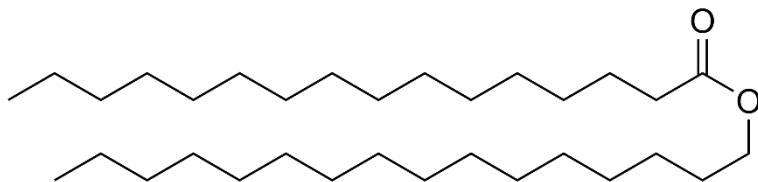
If the inside surface of each tube were coated with wax, would the general shape of the water meniscus change? Would the general shape of the mercury meniscus change?

Shape of Water Meniscus

- A. Yes, Inverted U
- B. No change
- C. No change
- D. Yes, Inverted U

Shape of Hg Meniscus

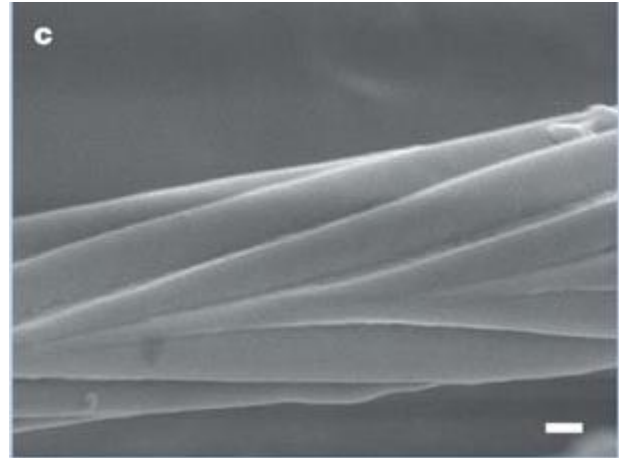
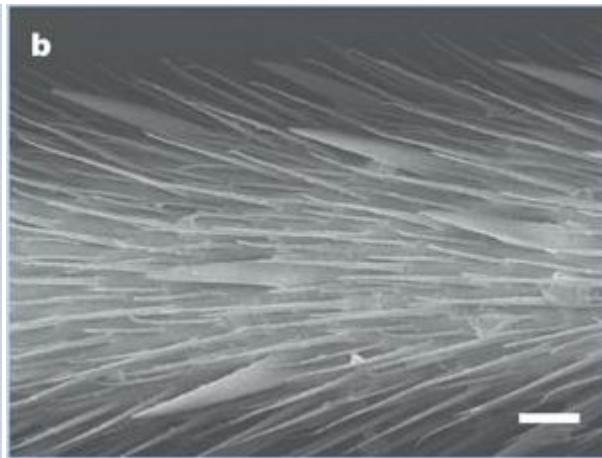
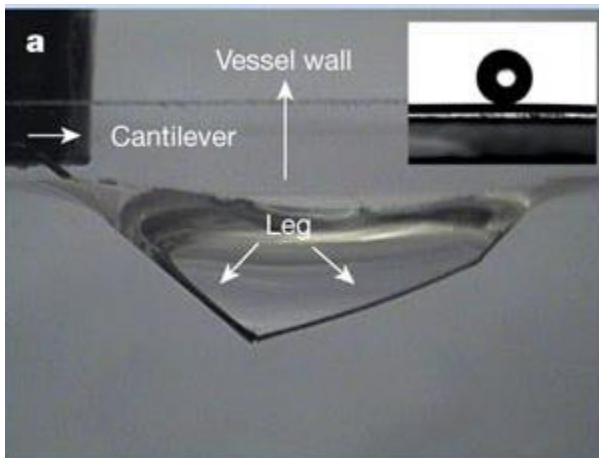
- Yes, downward U
- Yes, downward U
- No change
- No change



wax

Water-Repellent Legs of Water Striders

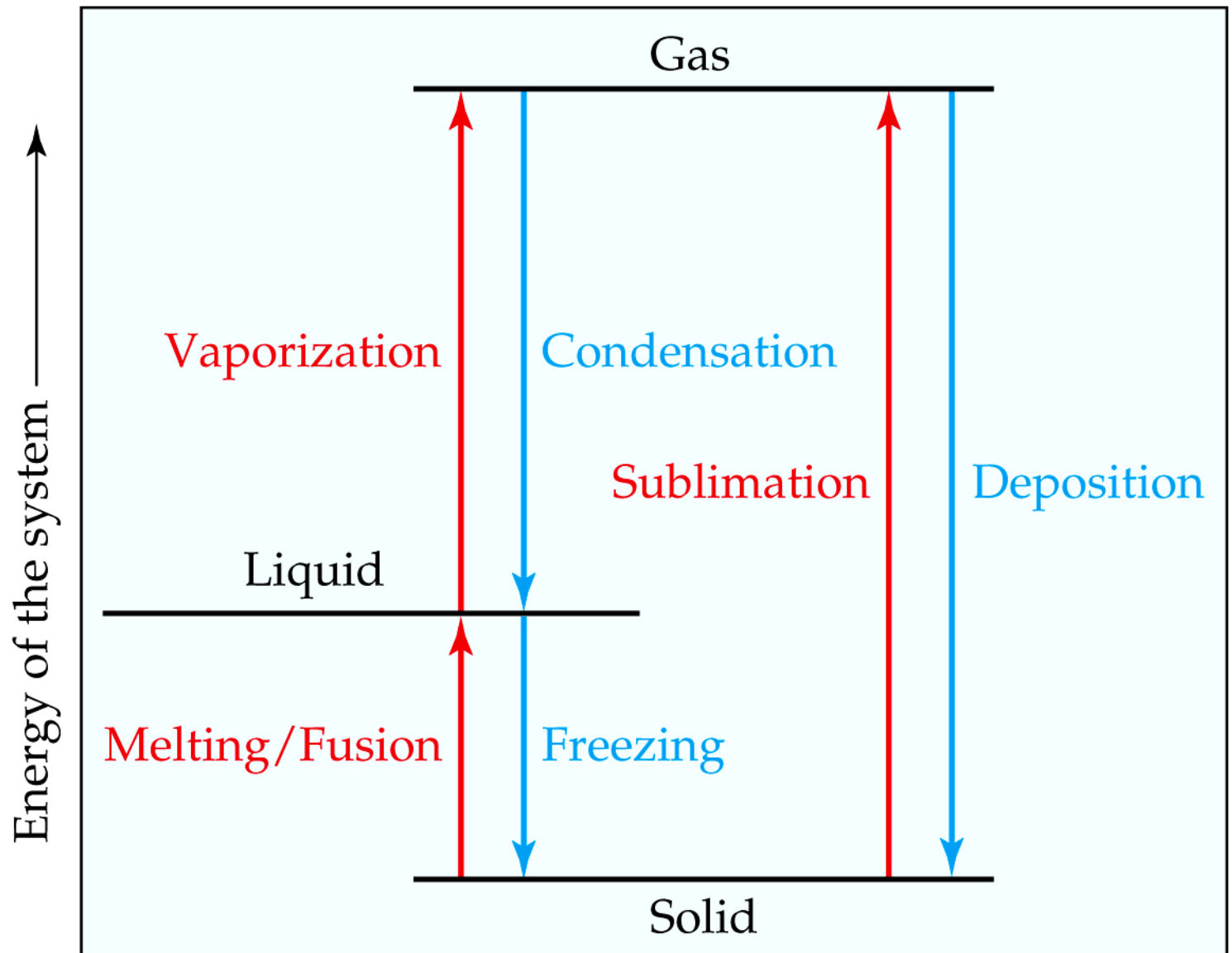
- Water striders (*Gerris remigis*) have remarkable non-wetting legs that enable them to stand effortlessly and move quickly on water, a feature believed to be due to a surface-tension effect caused by secreted wax.
- Jiang L and *et al.* show that it is the special hierarchical structure of the legs, which are covered by large numbers of oriented tiny hairs (microsetae) with fine nanogrooves, that is more important in inducing this water resistance.



The lotus leaf, super-hydrophobic, self-clean

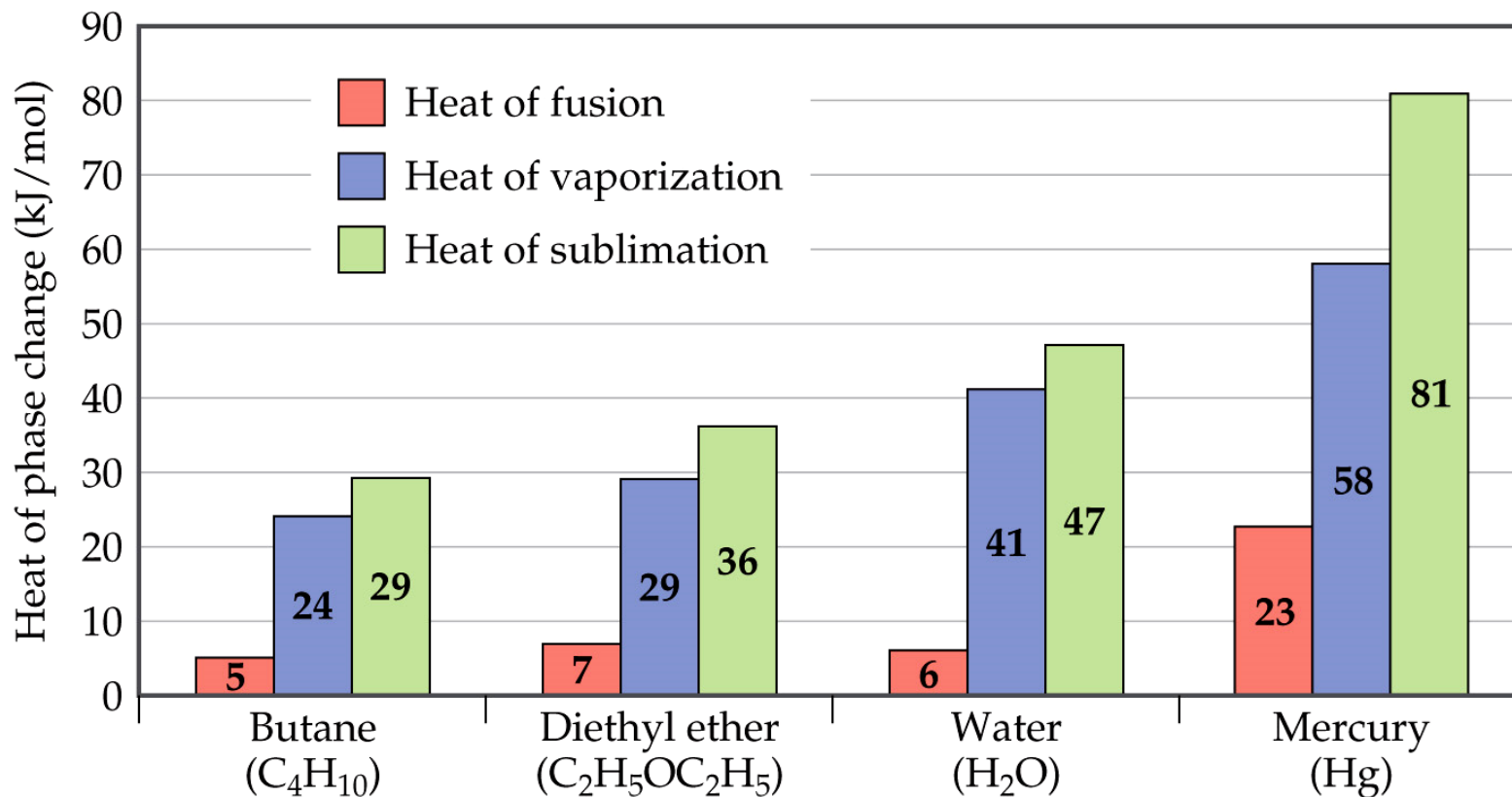


Phase Changes



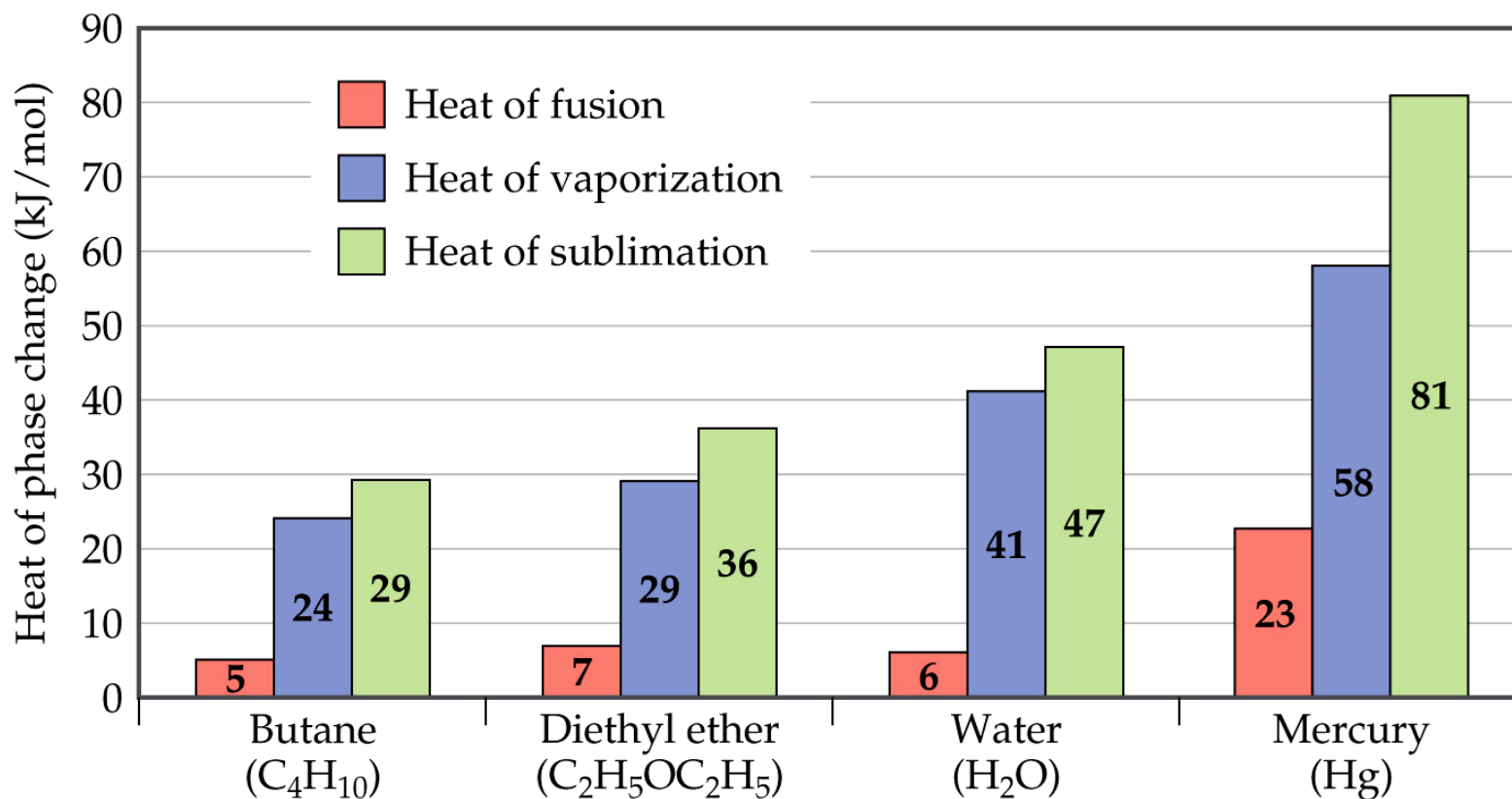
- Endothermic process (energy added to substance)
- Exothermic process (energy released from substance)

Energy Changes Associated with Changes of State



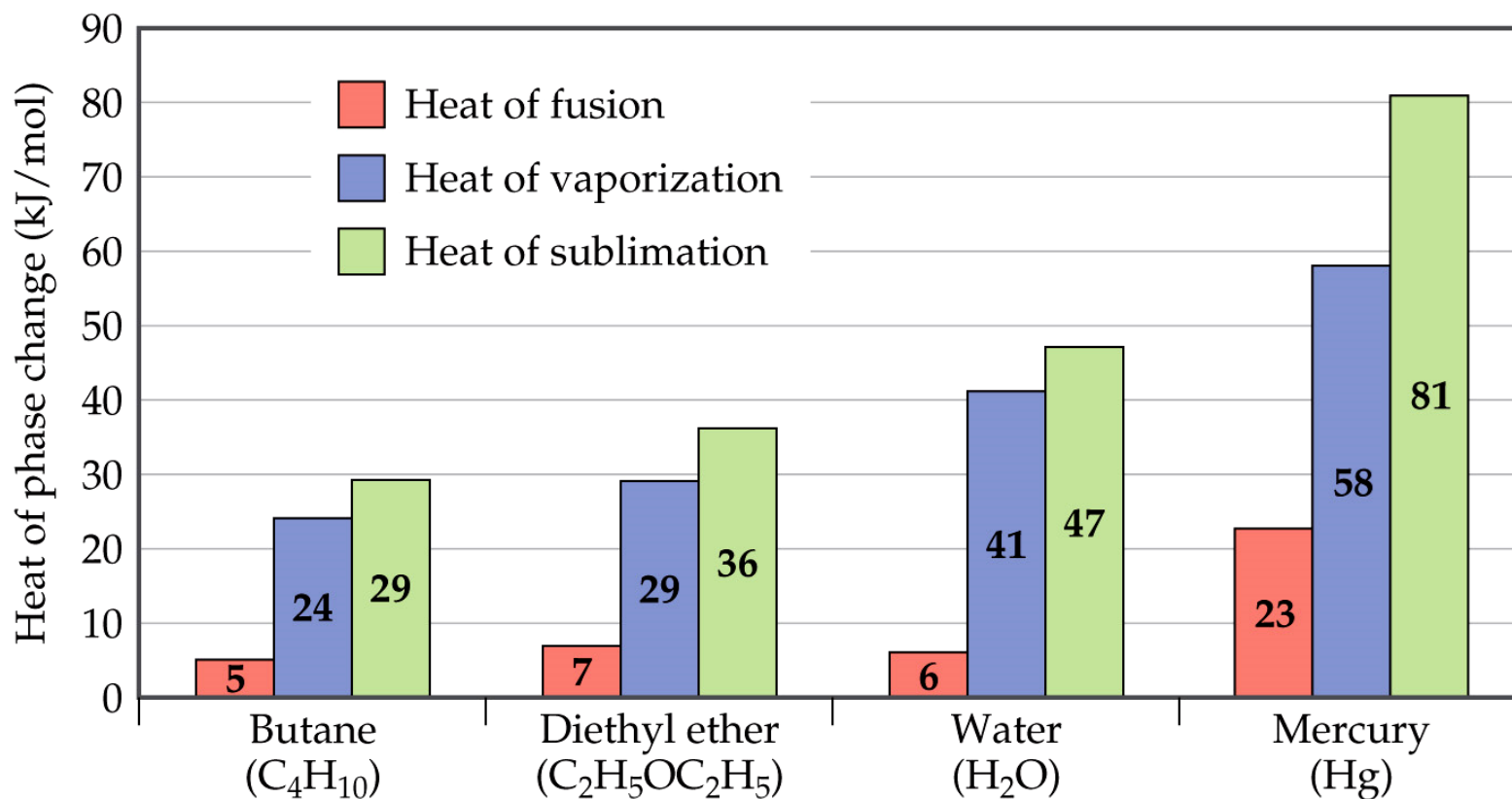
The **heat of fusion** is the energy required to change a solid at its melting point to a liquid.

Energy Changes Associated with Changes of State



The **heat of vaporization** is defined as the energy required to change a liquid at its boiling point to a gas.

Energy Changes Associated with Changes of State

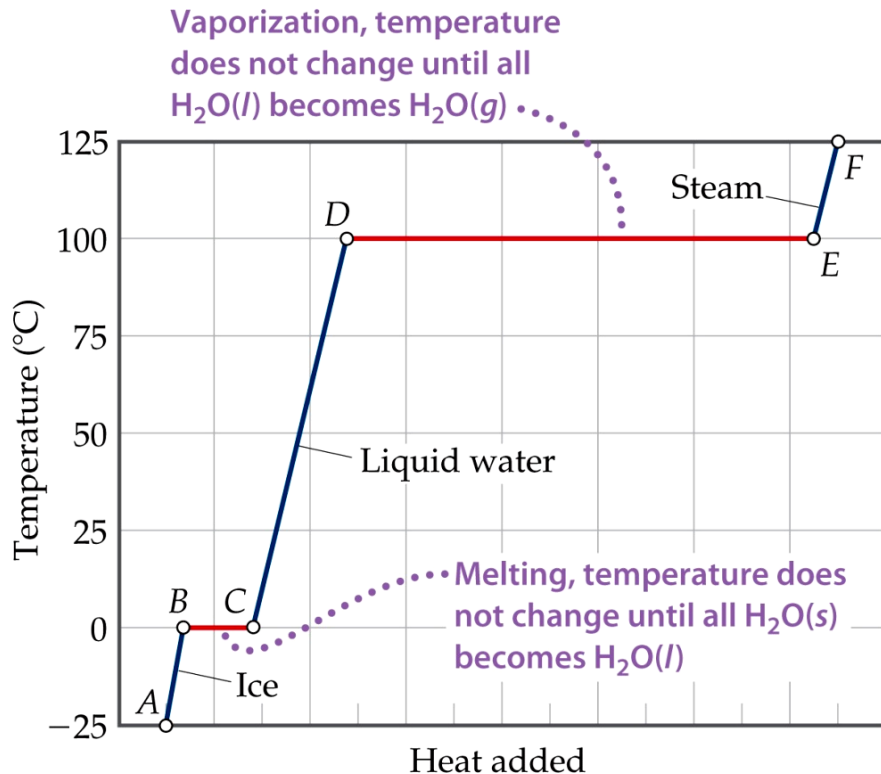


The **heat of sublimation** is defined as the energy required to change a solid directly to a gas.

Is it possible to calculate the heat of sublimation for a substance given its heats of vaporization and fusion? If so, what is the relationship?

- A. No, because we are not dealing with state functions.
- B. No, because we need heat of melting.
- C. Yes, $\Delta H_{sub} = \Delta H_{fus} + \Delta H_{vap}$
- D. Yes, $\Delta H_{sub} = \Delta H_{fus} - \Delta H_{vap}$

Energy Changes Associated with Changes of State



- The heat added to the system at the melting and boiling points goes into pulling the molecules farther apart from each other.
- The temperature of the substance does not rise during a phase change.

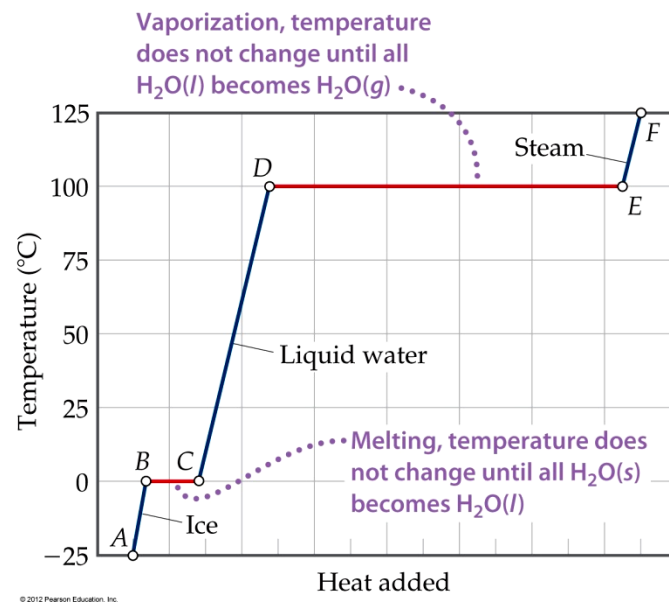
Exercise Calculating ΔH for Temperature and Phase Changes

Calculate the enthalpy change upon converting 1.00 mol of ice at $-25\text{ }^{\circ}\text{C}$ to steam at $125\text{ }^{\circ}\text{C}$ under a constant pressure of 1 atm. The specific heats of ice, liquid water, and steam are $2.03\text{ J/g}\cdot\text{K}$, $4.18\text{ J/g}\cdot\text{K}$, and $1.84\text{ J/g}\cdot\text{K}$, respectively. For H_2O , $\Delta H_{\text{fus}} = 6.01\text{ kJ/mol}$ and $\Delta H_{\text{vap}} = 40.67\text{ kJ/mol}$.

Solution

Plan We can calculate the enthalpy change for each segment and then sum them to get the total enthalpy change (Hess's law, Section 5.6).

Solve For segment AB in Figure 11.22, we are adding enough heat to ice to increase its temperature by $25\text{ }^{\circ}\text{C}$. A temperature change of $25\text{ }^{\circ}\text{C}$ is the same as a temperature change of 25 K , so we can use the specific heat of ice to calculate the enthalpy change during this process:



$$\text{AB: } \Delta H = (1.00\text{ mol})(18.0\text{ g/mol})(2.03\text{ J/g}\cdot\text{K})(25\text{ K}) = 914\text{ J} = 0.91\text{ kJ}$$

Exercise Calculating ΔH for Temperature and Phase Changes (continued)

For segment BC in Figure 11.22, in which we convert ice to water at $0\text{ }^{\circ}\text{C}$, we can use the molar enthalpy of fusion directly:

$$BC: \Delta H = (1.00\text{ mol})(6.01\text{ kJ/mol}) = 6.01\text{ kJ}$$

The enthalpy changes for segments CD , DE , and EF can be calculated in similar fashion:

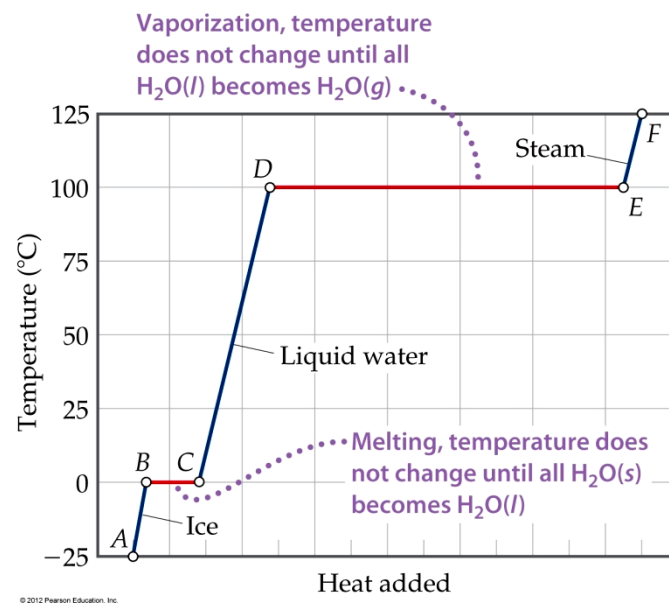
$$CD: \Delta H = (1.00\text{ mol})(18.0\text{ g/mol})(4.18\text{ J/g}\cdot\text{K})(100\text{ K}) = 7520\text{ J} = 7.52\text{ kJ}$$

$$DE: \Delta H = (1.00\text{ mol})(40.67\text{ kJ/mol}) = 40.7\text{ kJ}$$

$$EF: \Delta H = (1.00\text{ mol})(18.0\text{ g/mol})(1.84\text{ J/g}\cdot\text{K})(25\text{ K}) = 830\text{ J} = 0.83\text{ kJ}$$

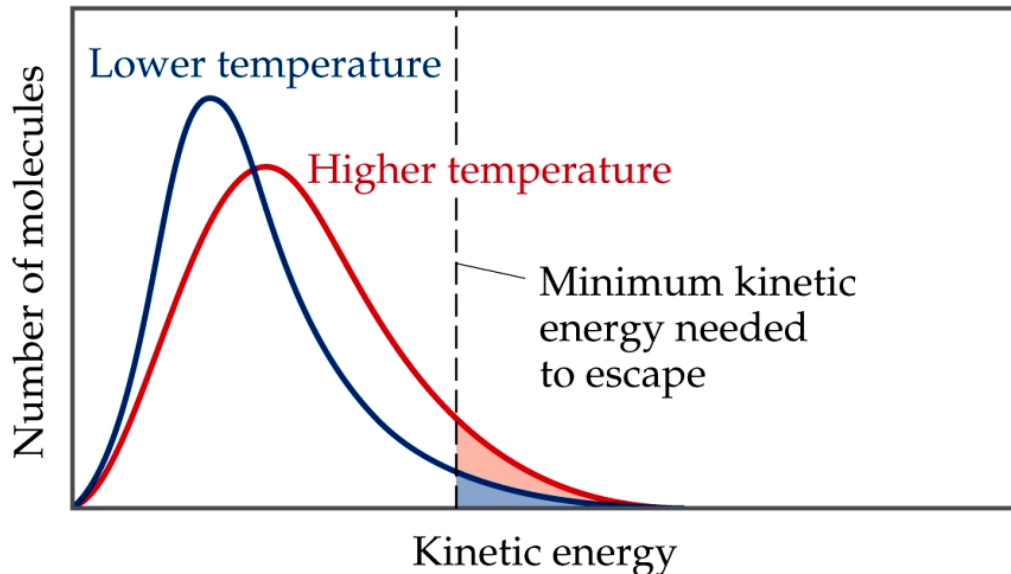
The total enthalpy change is the sum of the changes of the individual steps:

$$\Delta H = 0.91\text{ kJ} + 6.01\text{ kJ} + 7.52\text{ kJ} + 40.7\text{ kJ} + 0.83\text{ kJ} = 56.0\text{ kJ}$$



Vapor Pressure

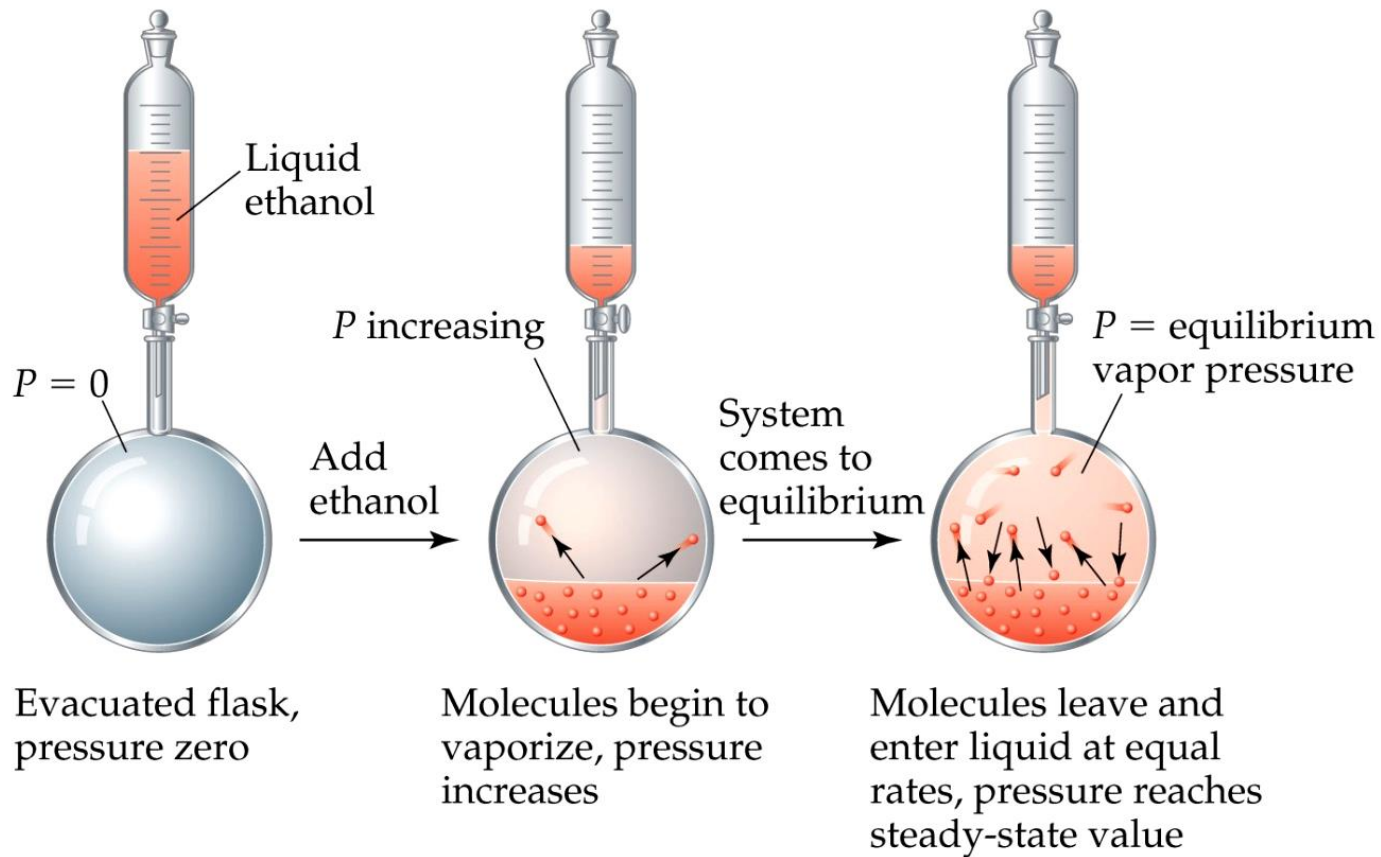
- At any temperature some molecules in a liquid have enough energy to break free.
- As the temperature rises, the fraction of molecules that have enough energy to break free increases.



Blue area = number of molecules having enough energy to evaporate at lower temperature

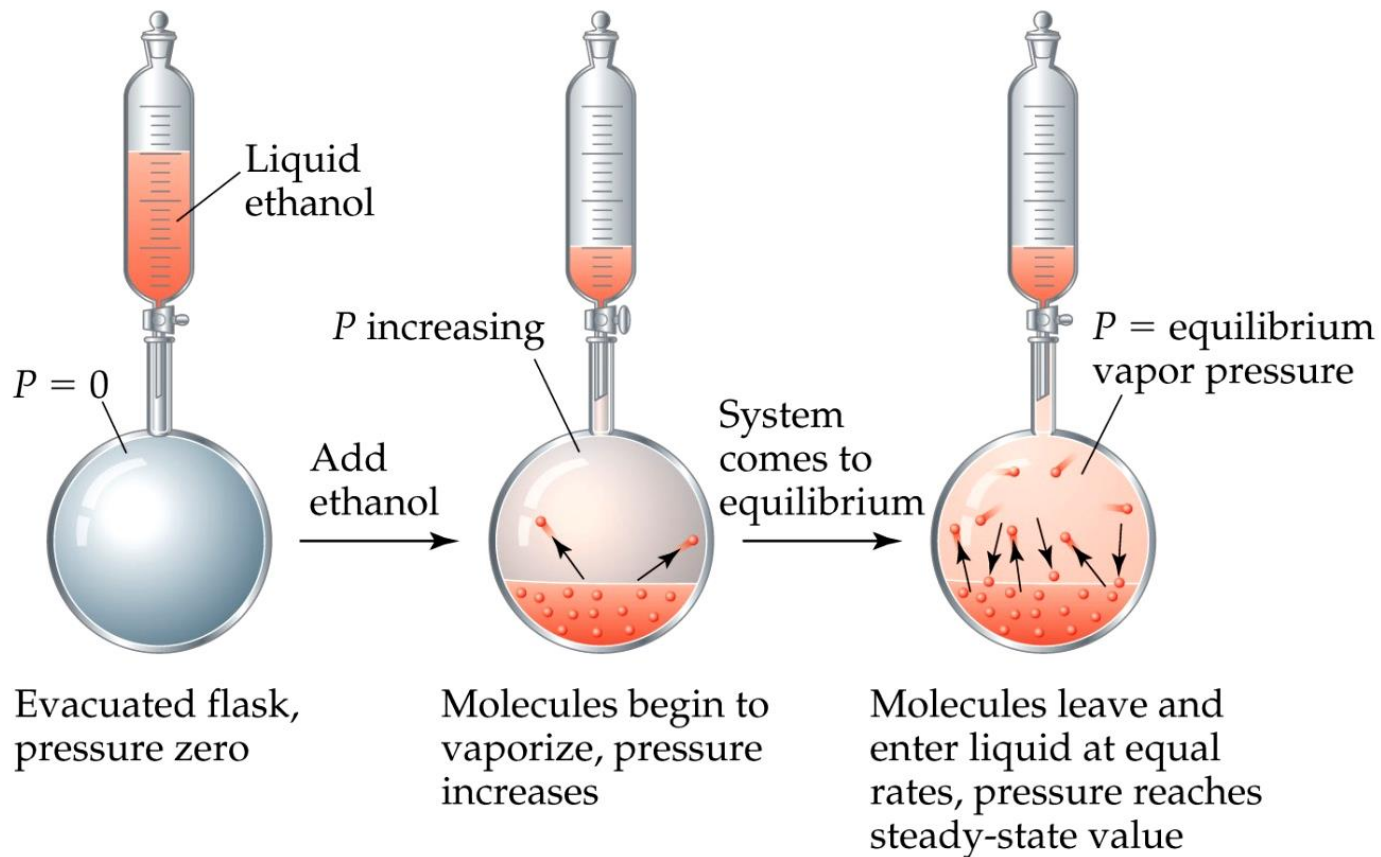
Red + blue areas = number of molecules having enough energy to evaporate at higher temperature

Vapor Pressure



As more molecules escape the liquid, the pressure they exert increases.

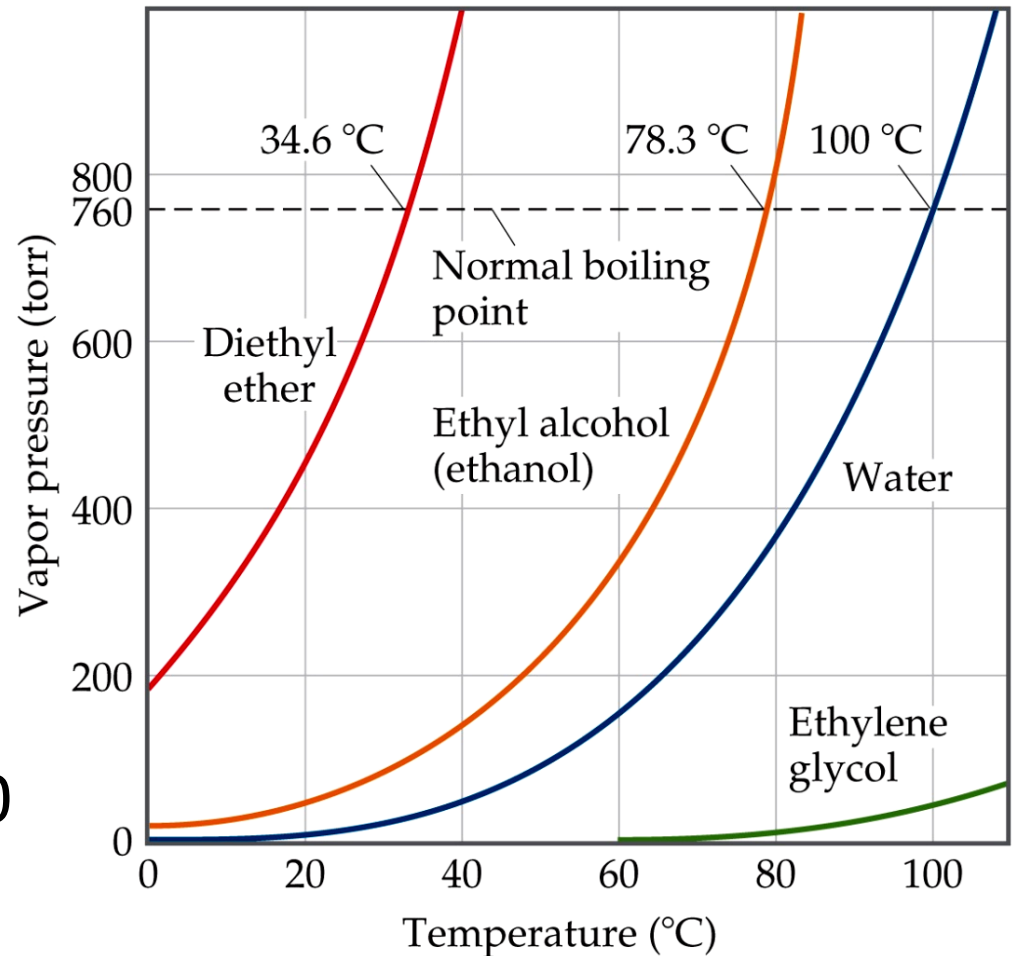
Vapor Pressure



The liquid and vapor reach a state of dynamic equilibrium: liquid molecules evaporate and vapor molecules condense *at the same rate*.

Vapor Pressure

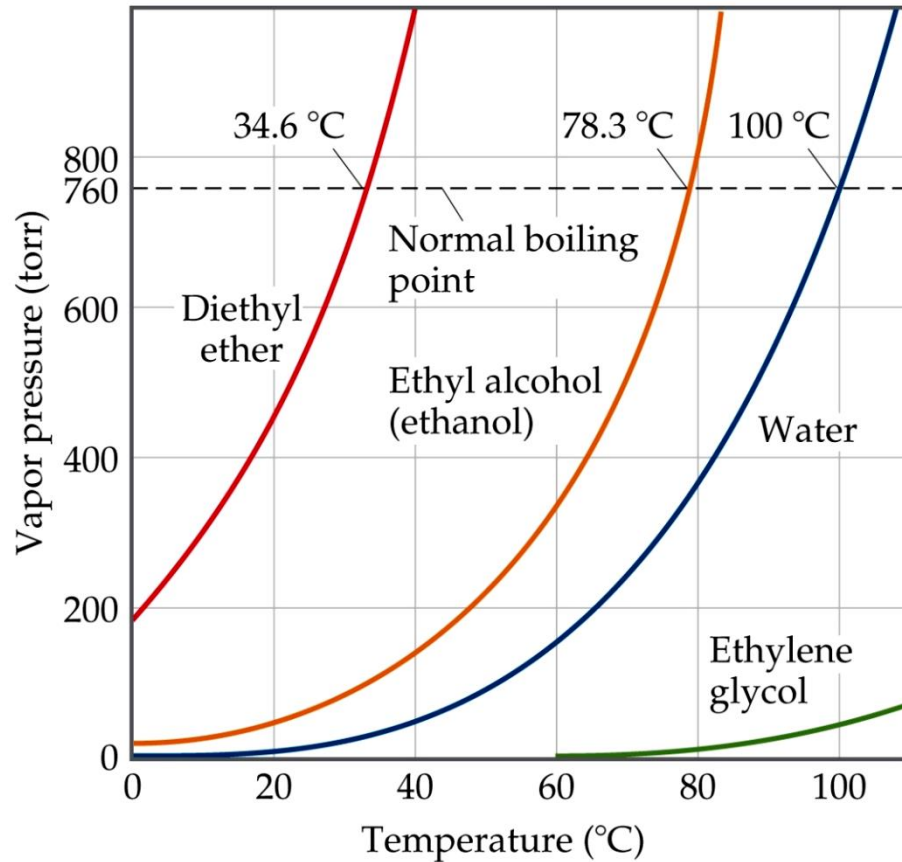
- The **boiling point** of a liquid is the temperature at which its vapor pressure equals atmospheric pressure.
- The **normal boiling point** is the temperature at which its vapor pressure is 760 torr.



760 torr. = 101325 pa

Vapor Pressure

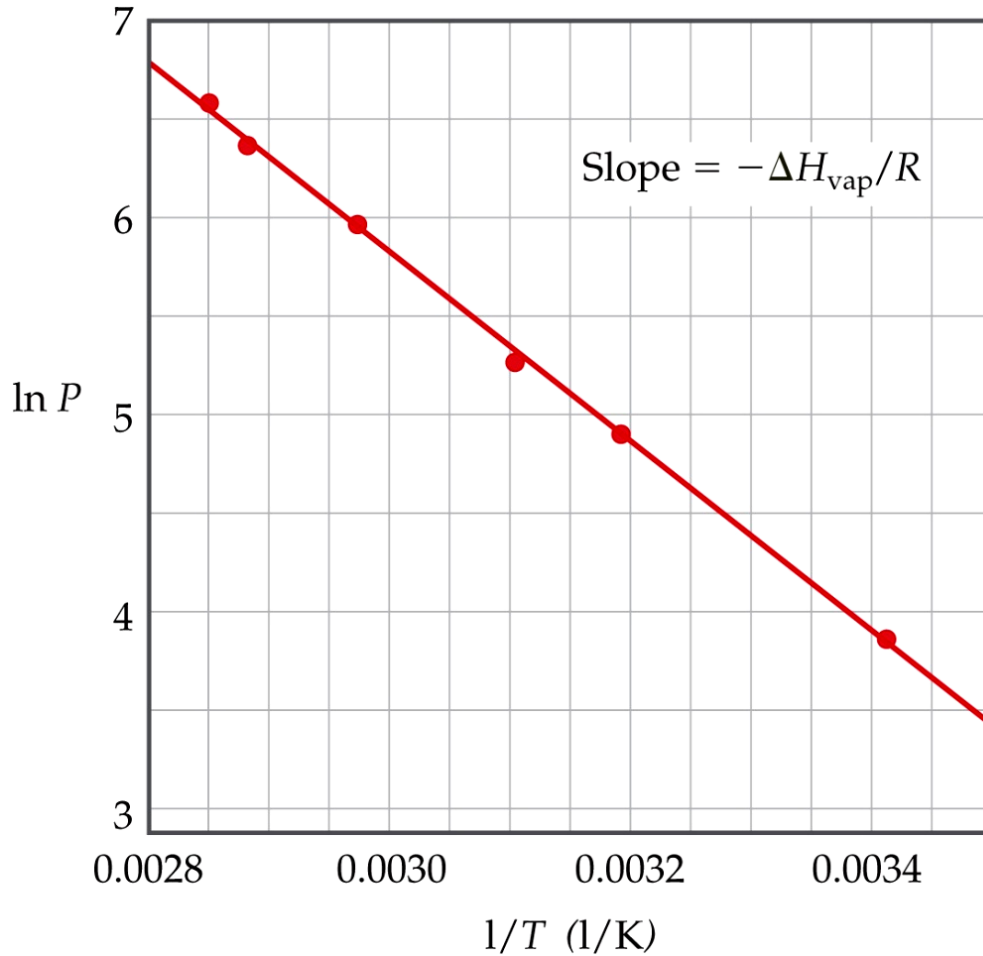
high pressure, high temperature of boiling point



low pressure, low temperature of boiling point

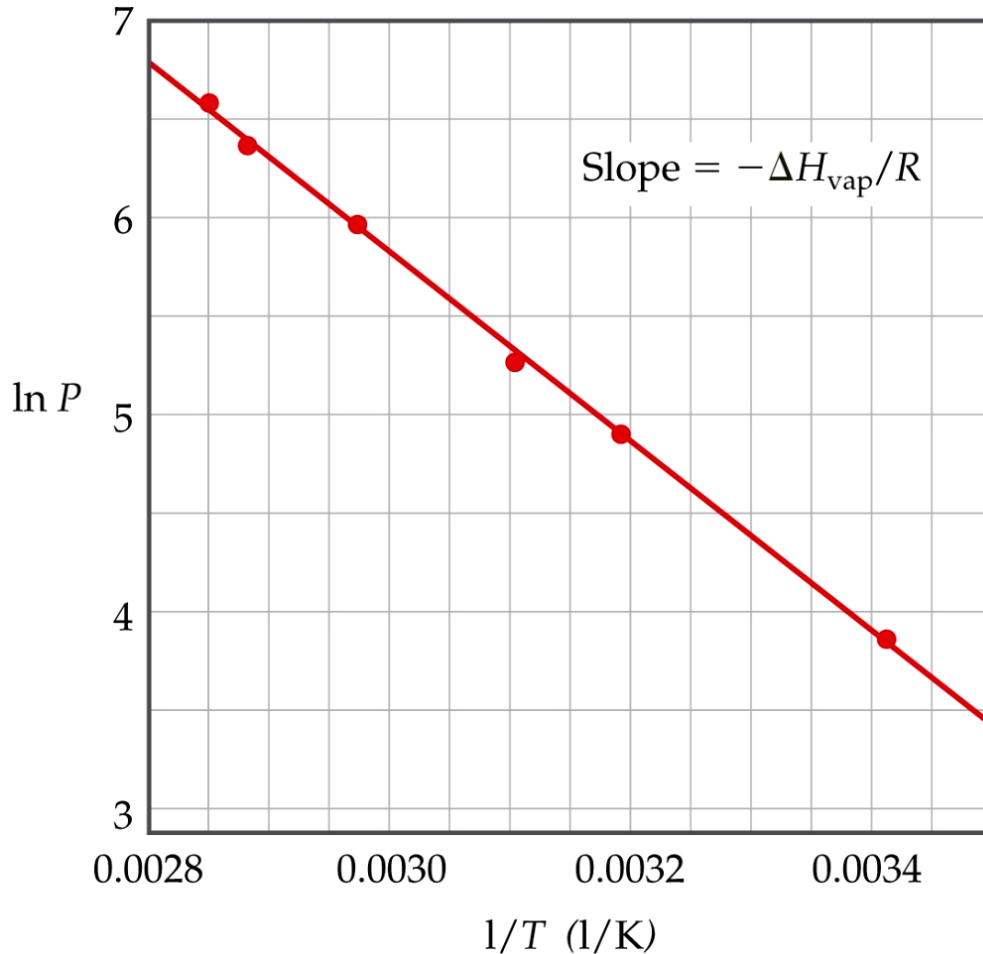


Vapor Pressure



The natural log of the vapor pressure of a liquid is inversely proportional to its temperature.

Vapor Pressure



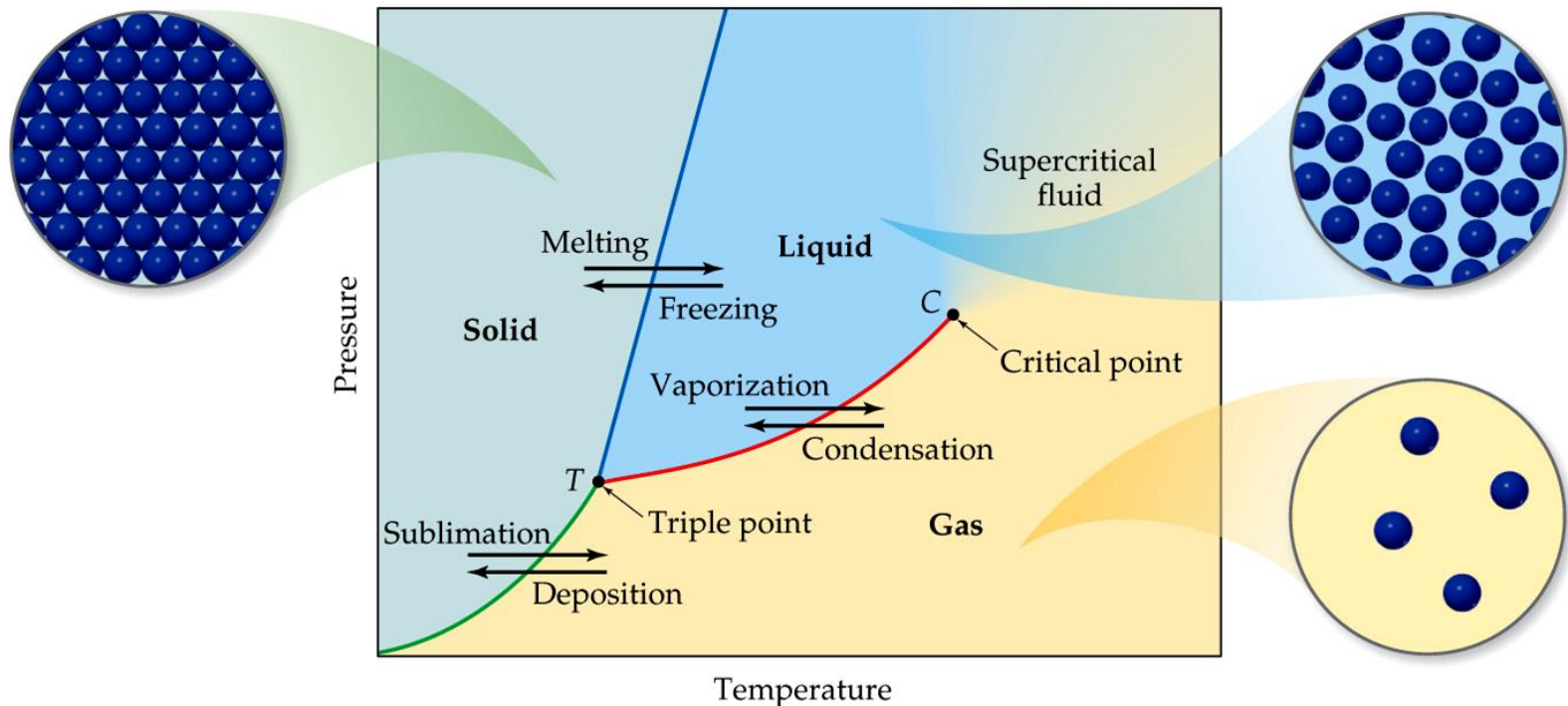
This relationship is quantified in the **Clausius–Clapeyron** equation:

$$\ln P = -\Delta H_{\text{vap}}/RT + C,$$

where C is a constant.

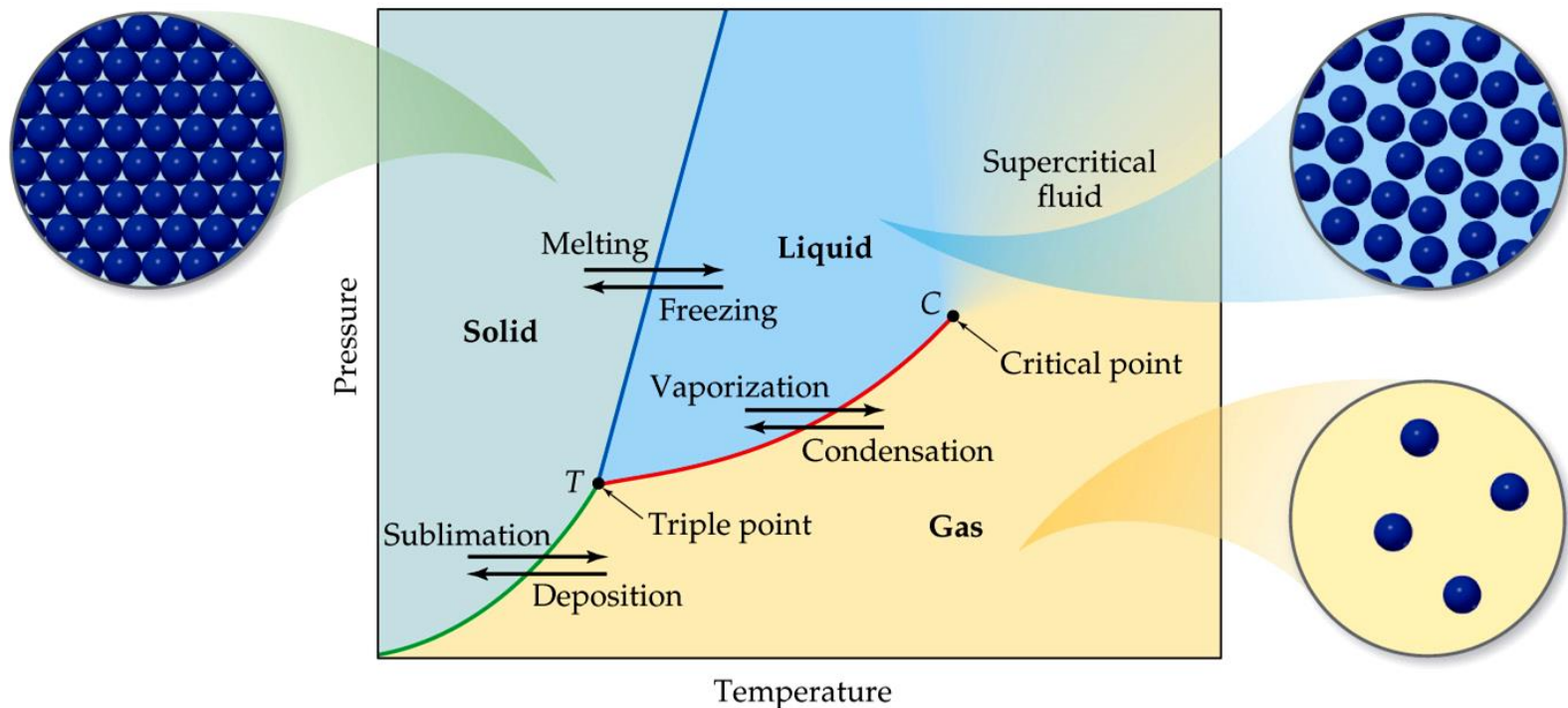
Phase Diagrams

Phase diagrams display the state of a substance at various pressures and temperatures, and the places where equilibria exist between phases.



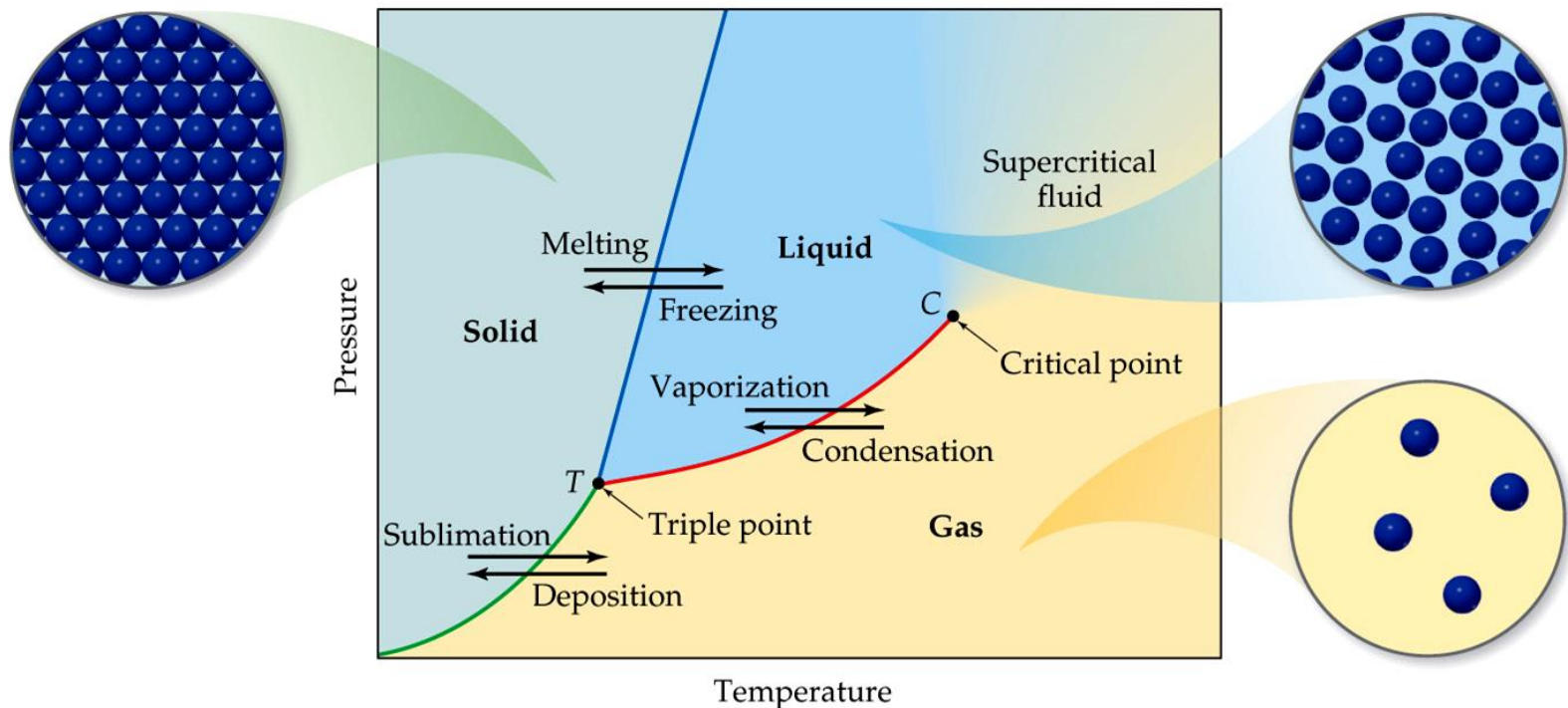
Phase Diagrams

The liquid–vapor interface starts at the triple point (T), at which all three states are in equilibrium, and ends at the critical point (C), above which the liquid and vapor are indistinguishable from each other.



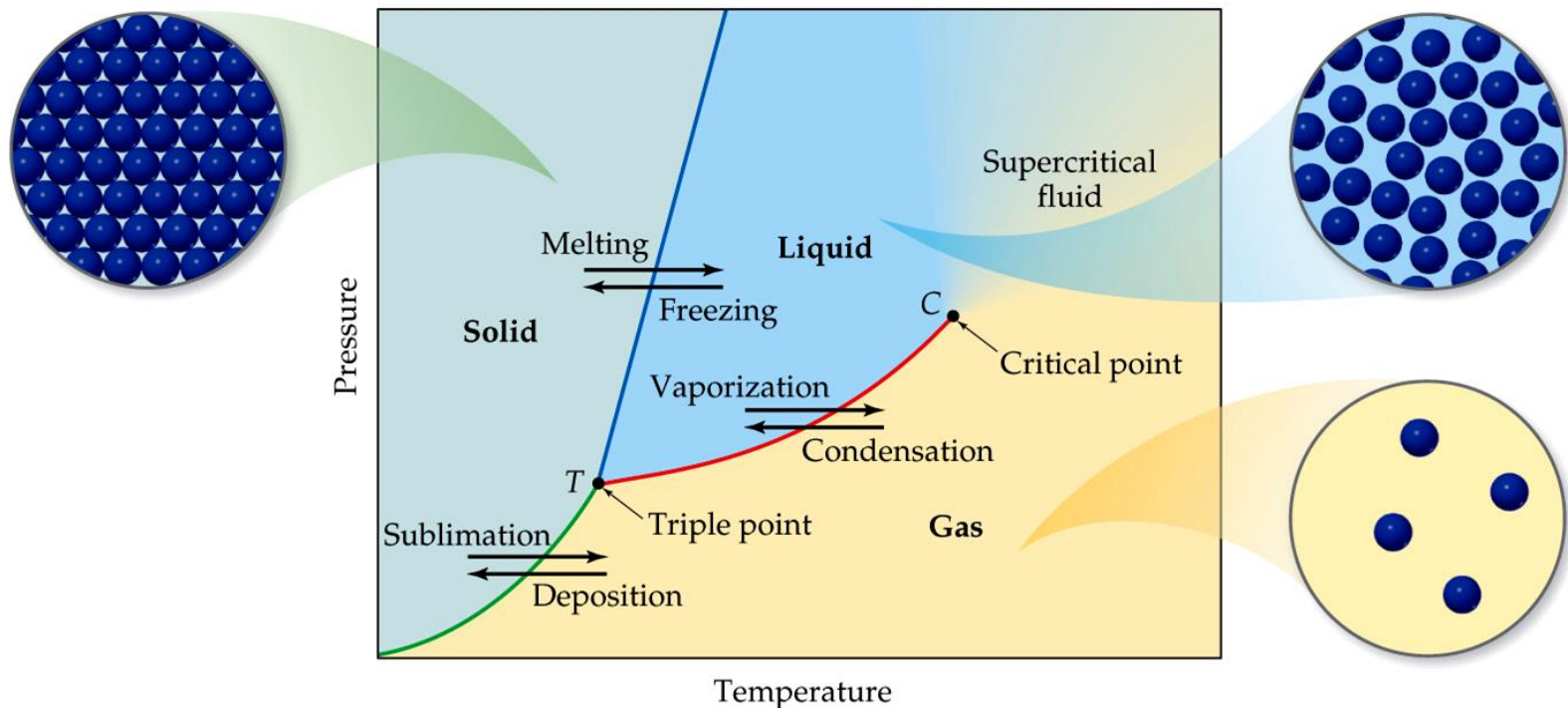
Phase Diagrams

Each point along this line is the boiling point of the substance at that pressure.



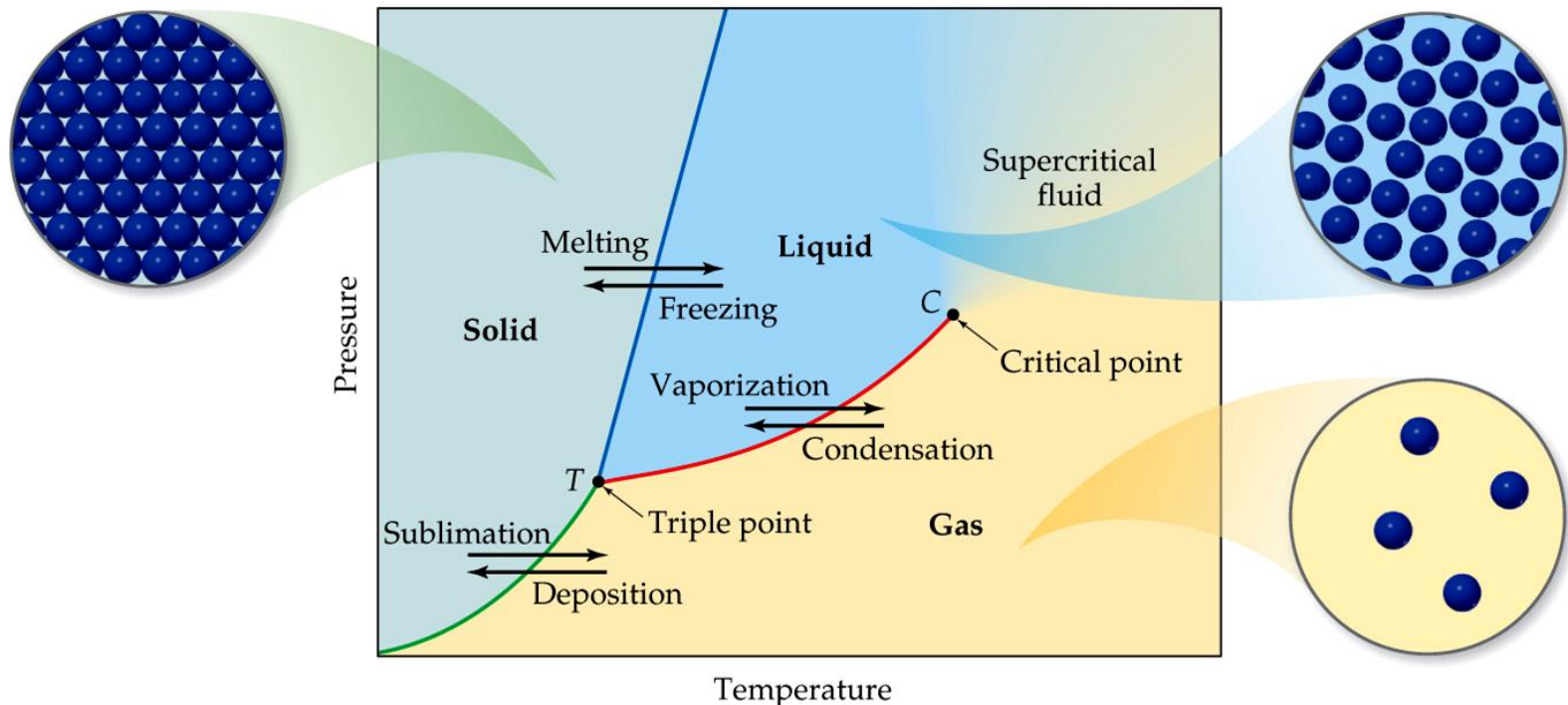
Phase Diagrams

The interface between liquid and solid marks the melting point of a substance at each pressure.



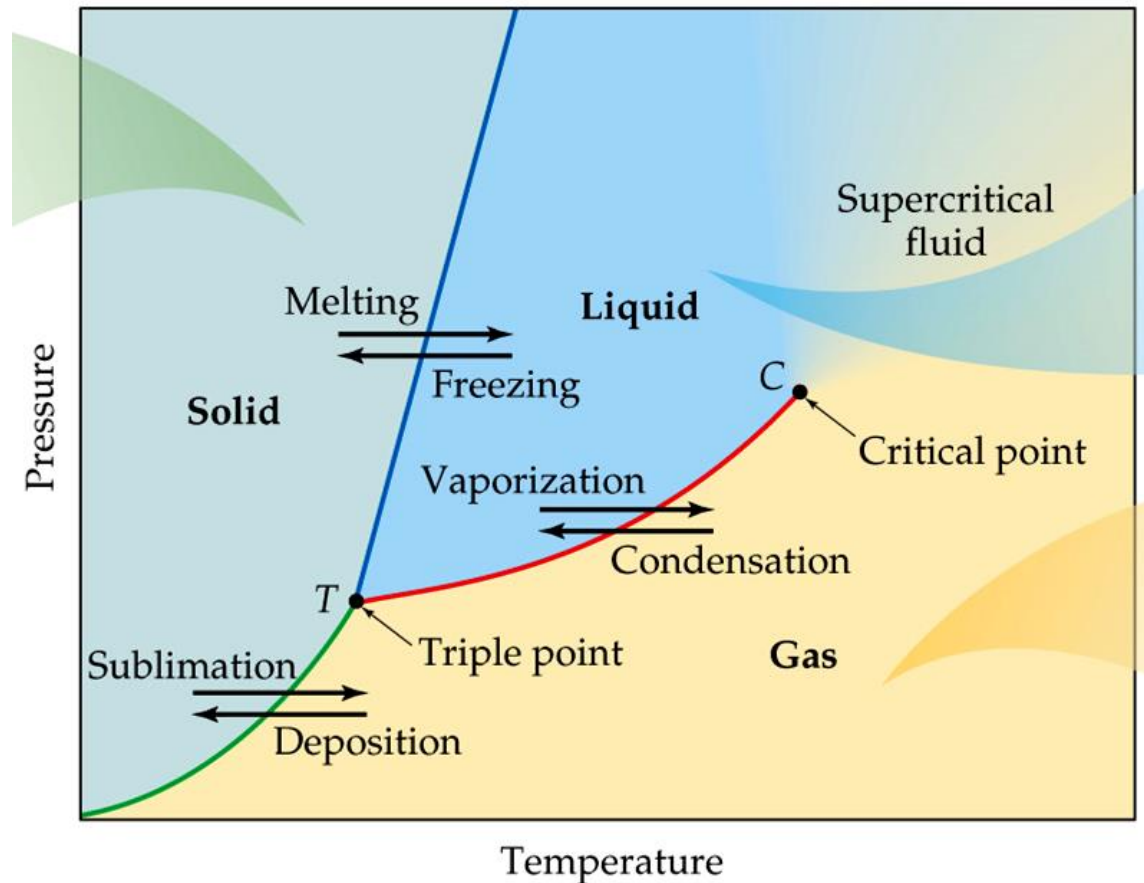
Phase Diagrams

- Below the triple point the substance cannot exist in the liquid state.
- Along the solid–gas line those two phases are in equilibrium; the sublimation point at each pressure is along this line.

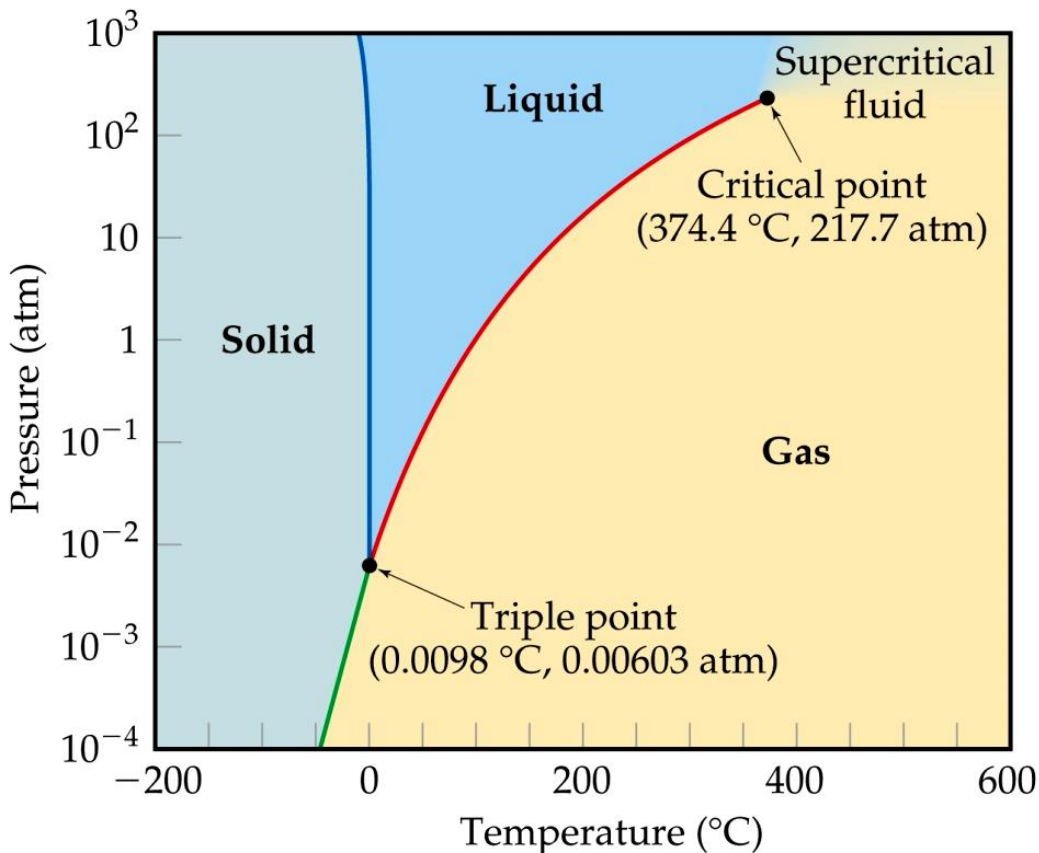


If the pressure exerted on a liquid is increased, while the temperature is held constant, what type of phase transition will eventually occur?

- A. Freezing
- B. Melting
- C. Vaporization
- D. Condensation

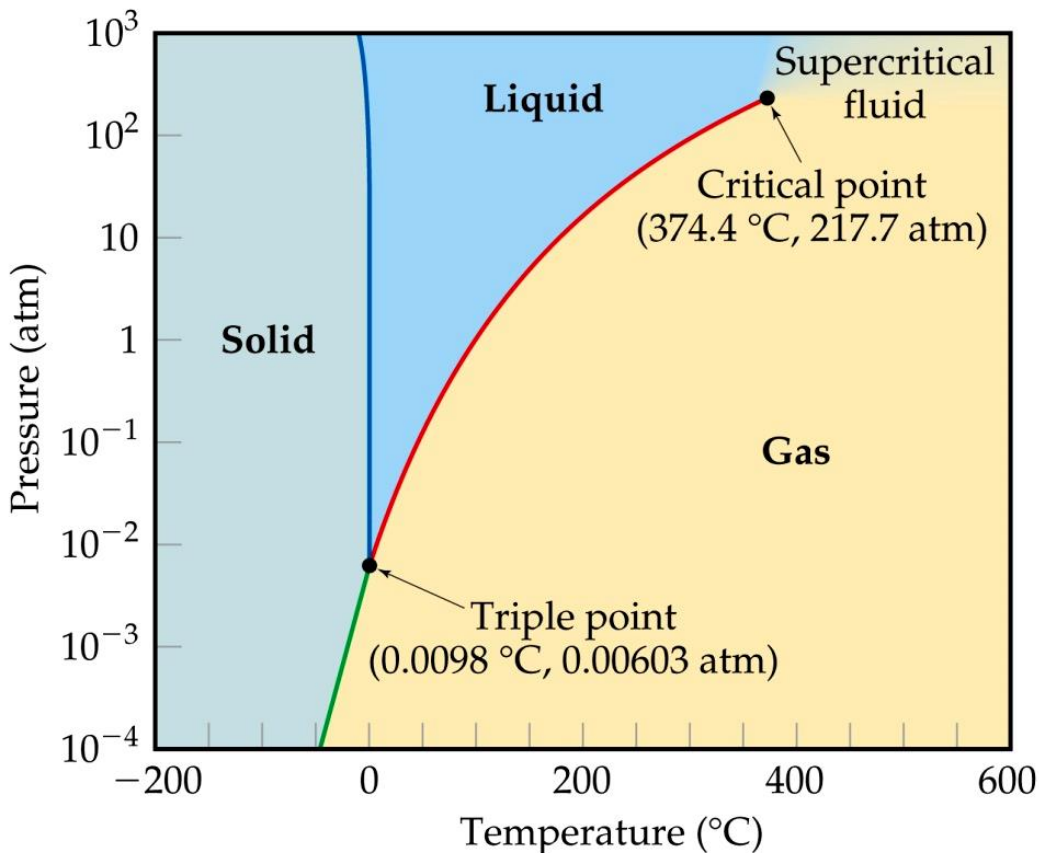


Phase Diagram of Water



- Note the high critical temperature and critical pressure.
 - These are due to the strong van der Waals forces between water molecules.

Phase Diagram of Water

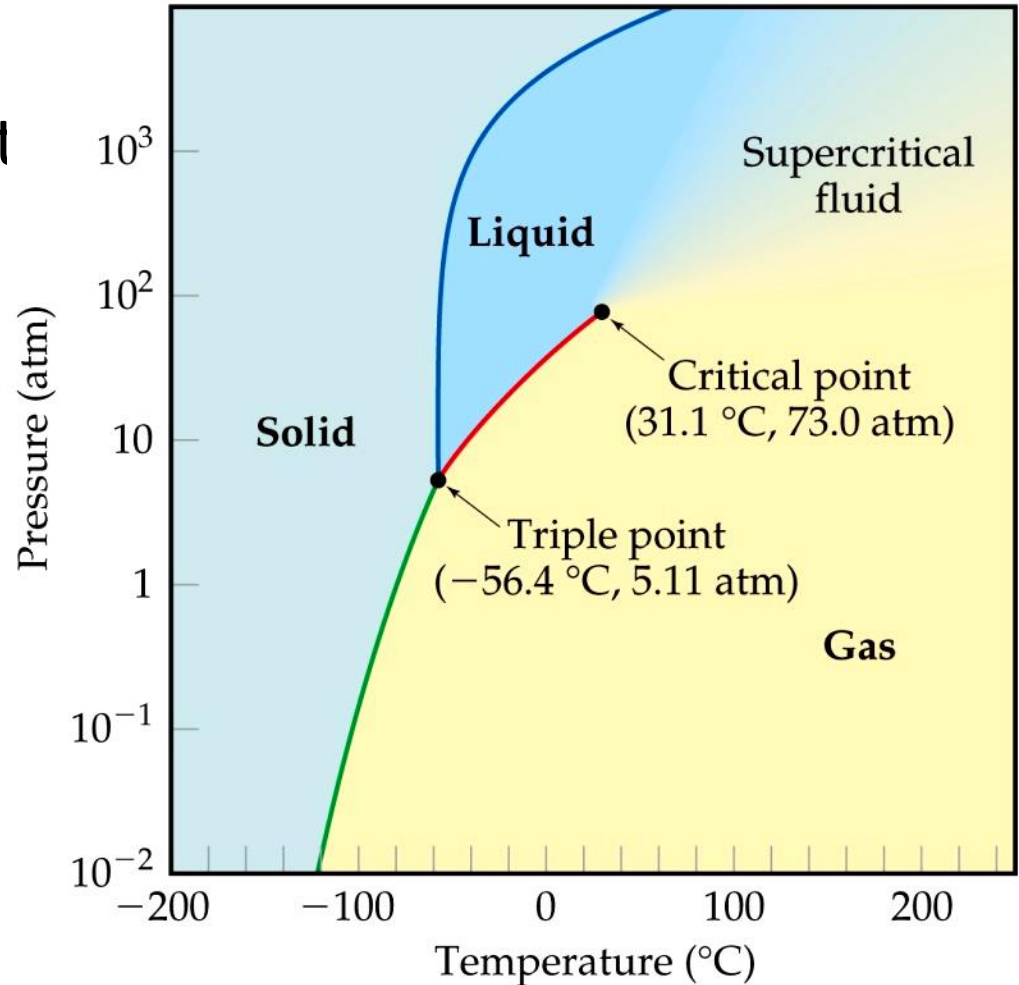
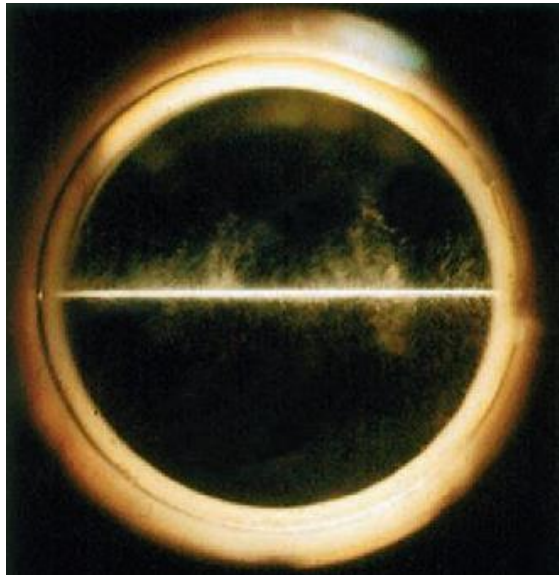


The slope of the solid–liquid line is negative.

- This means that as the pressure is increased at a temperature just below the melting point, water goes from a solid to a liquid.

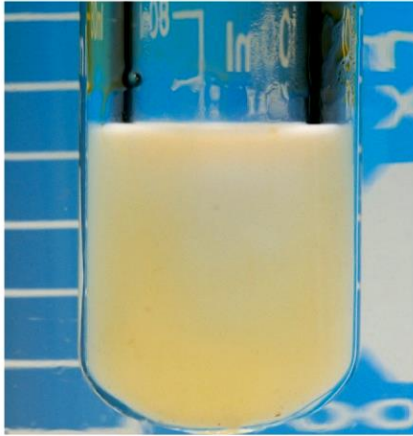
Phase Diagram of Carbon Dioxide

Carbon dioxide cannot exist in the liquid state at pressures below 5.11 atm; CO₂ sublimates at normal pressures.

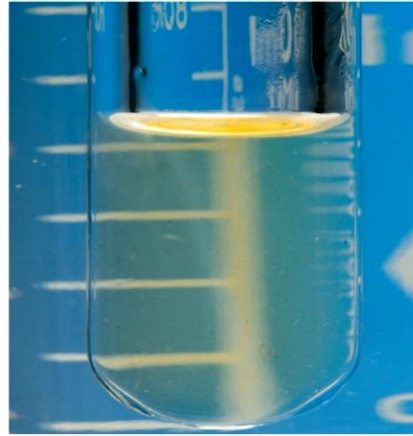


Supercritical Fluid of CO₂

Liquid Crystals

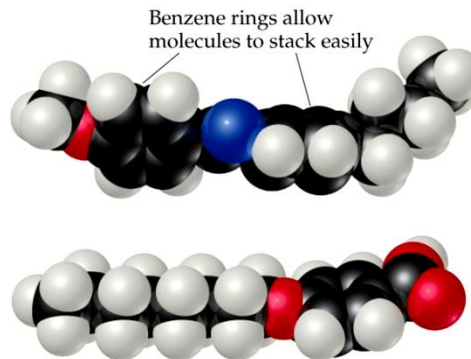
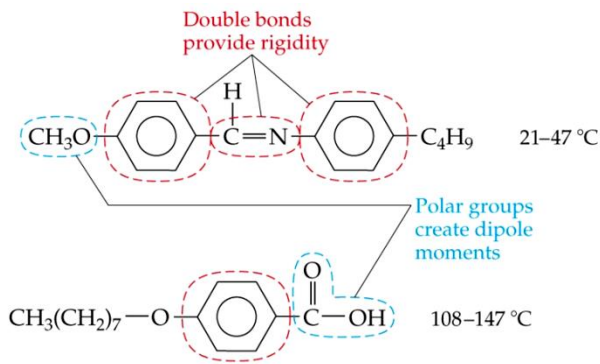


$145\text{ }^{\circ}\text{C} < T < 179\text{ }^{\circ}\text{C}$
Liquid crystalline phase

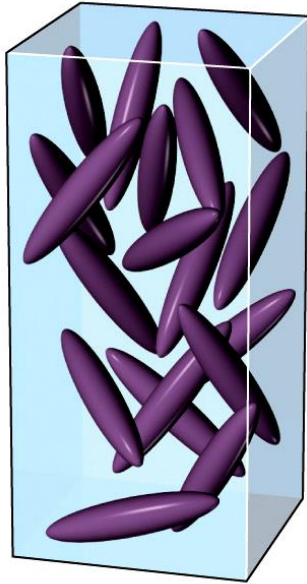


$T > 179\text{ }^{\circ}\text{C}$
Liquid phase

- Some substances do not go directly from the solid state to the liquid state.
- In this intermediate state, liquid crystals have some traits of solids and some of liquids.

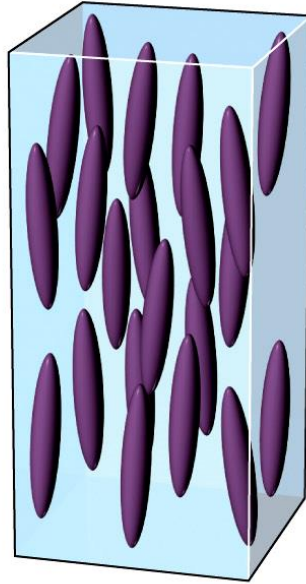


Liquid Crystals



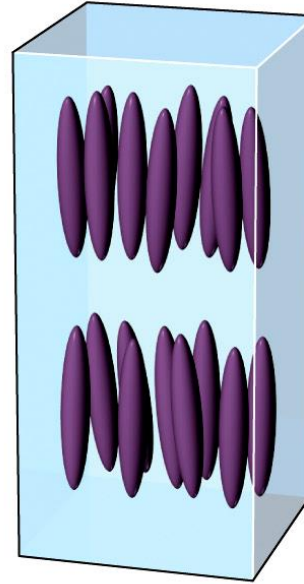
Liquid phase

Molecules
arranged
randomly



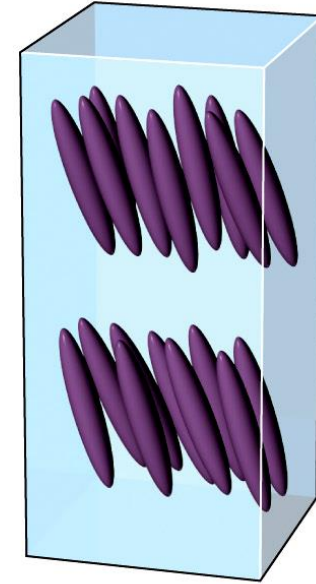
**Nematic liquid
crystalline phase**

Long axes of
molecules
aligned, but ends
are not aligned



**Smectic A liquid
crystalline phase**

Molecules aligned
in layers, long
axes of molecules
perpendicular to
layer planes

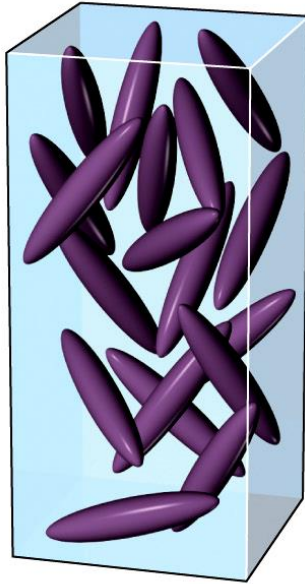


**Smectic C liquid
crystalline phase**

Molecules aligned
in layers, long
axes of molecules
inclined with
respect to layer
planes

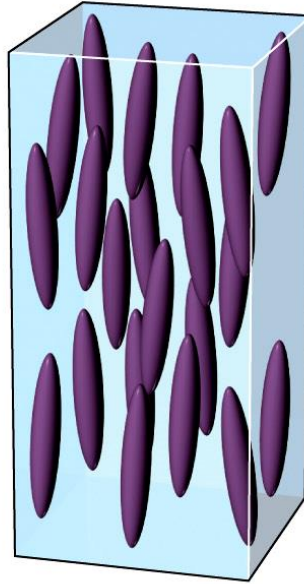
Unlike liquids, molecules in liquid crystals have some degree of order.

Liquid Crystals



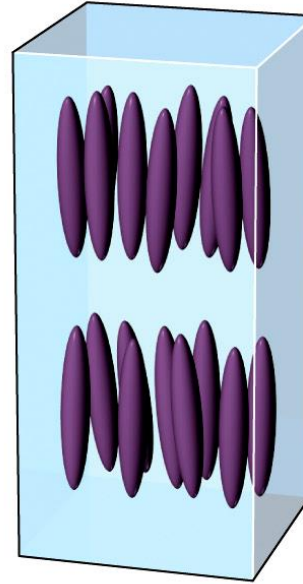
Liquid phase

Molecules
arranged
randomly



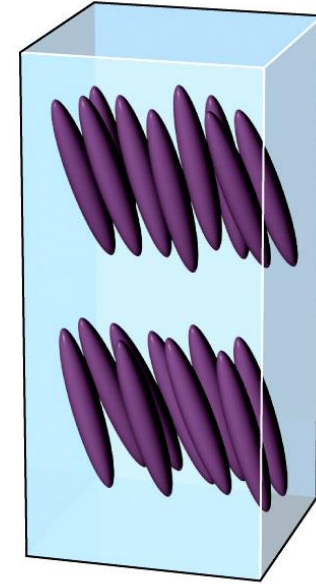
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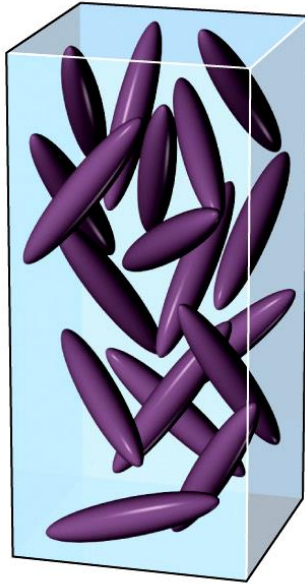


**Smectic C liquid
crystalline phase**

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in layers, long
axes of molecules
inclined with
respect to layer
planes

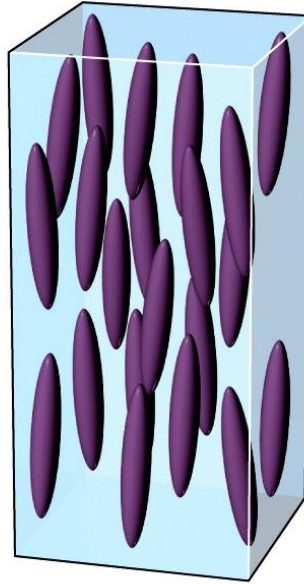
In nematic liquid crystals, molecules are only ordered in one dimension, along the long axis.

Liquid Crystals



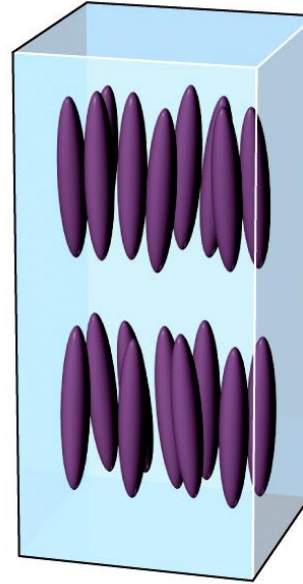
Liquid phase

Molecules
arranged
randomly



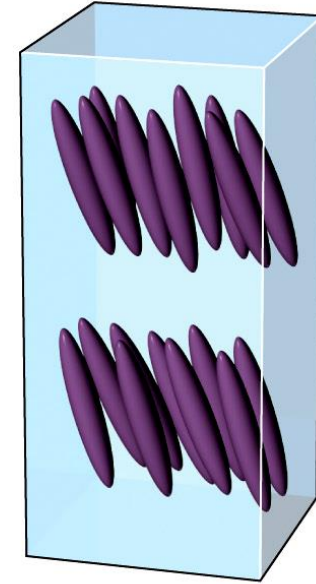
**Nematic liquid
crystalline phase**

Long axes of
molecules
aligned, but ends
are not aligned



**Smectic A liquid
crystalline phase**

Molecules aligned
in layers, long
axes of molecules
perpendicular to
layer planes

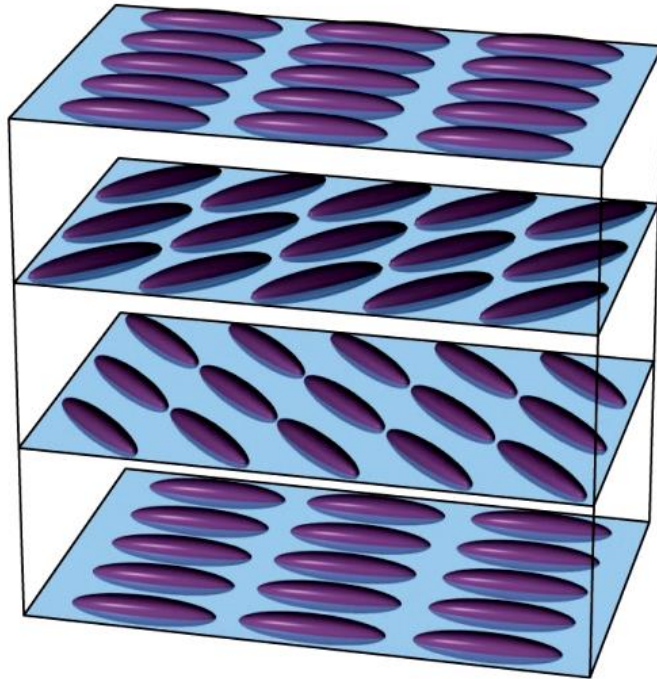


**Smectic C liquid
crystalline phase**

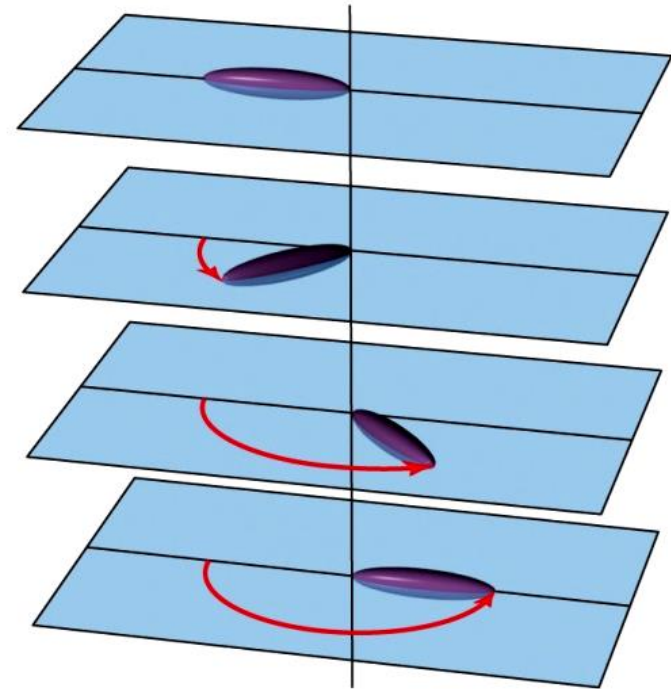
Molecules aligned
in layers, long
axes of molecules
inclined with
respect to layer
planes

In smectic liquid crystals, molecules are ordered in two dimensions, along the long axis and in layers.

Liquid Crystals



In a cholesteric liquid crystal the molecules pack into layers; the long axis of each molecule is oriented parallel to its neighbors within the same layer



The direction along which the molecules point rotates from one layer to the next, resulting in a spiraling pattern resembling the threads of a screw

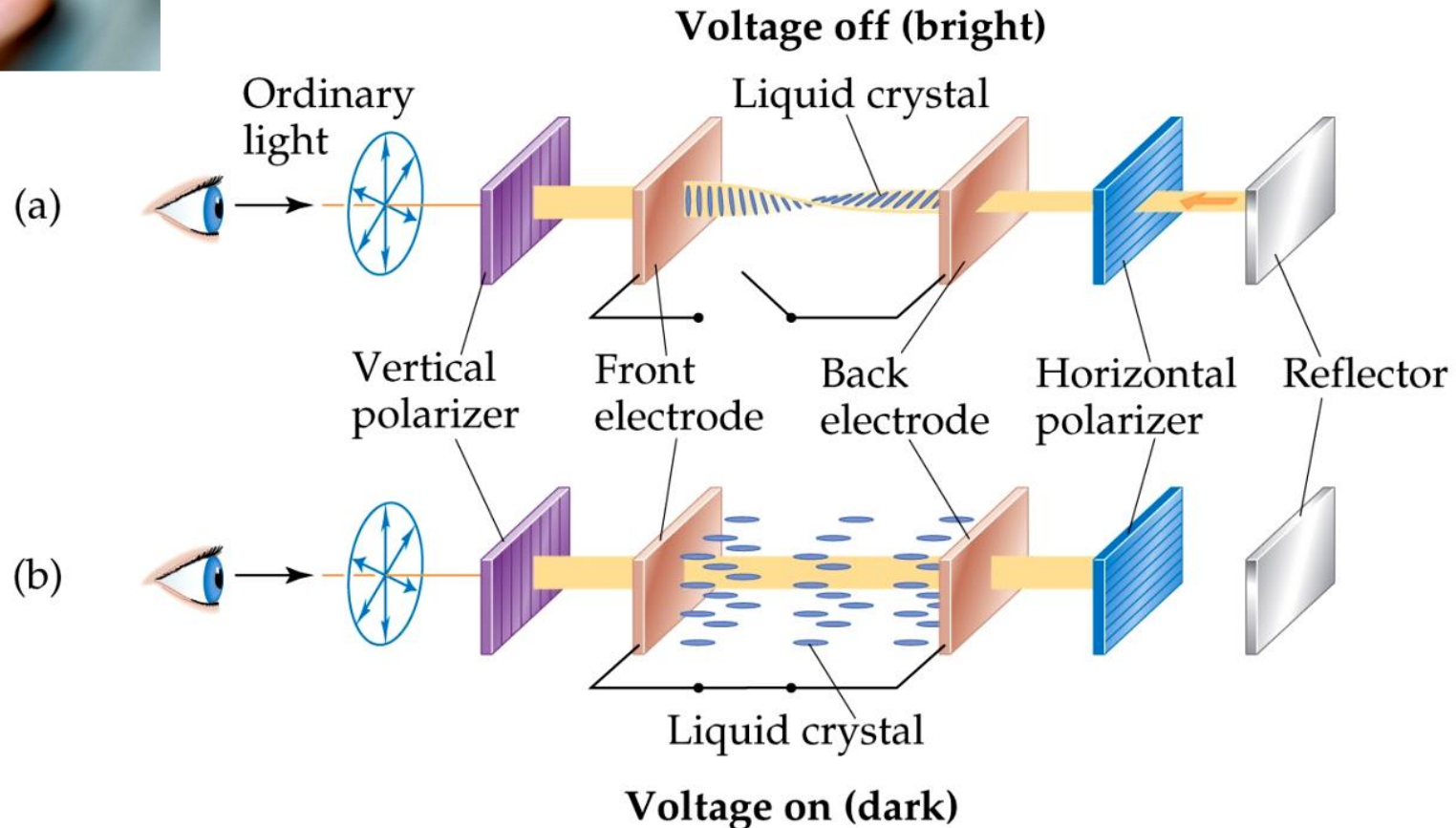
In **cholesteryl liquid crystals**, nematic-like crystals are layered at angles to each other.



Liquid Crystals

These crystals can exhibit color changes with changes in temperature and pressure.

electronic
paper



The Beauty of Liquid Crystals



Ionic Liquids - Green Solvents for the Future

- Salts that are liquid at ambient temperatures.
- Low melting organic salts
- Very low vapor pressure
- Non-flammable
- Wide liquid range (300 °C)
- Moderate to high viscosity (bad)
- Solvent properties different from molecular solvents
- Solvent properties can be controlled by selection of anion and cation – hence often termed designer solvents (good).

Ionic Liquids - Green Solvents for the Future

Alkylammonium
Alkylphosphonium
Alkylnitride
N,N'-dialkylimidazolium
N-alkylpyridinium

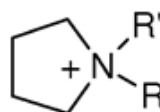
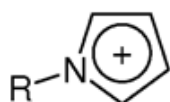
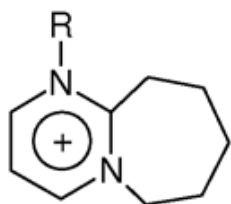
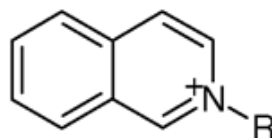
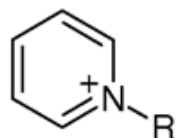
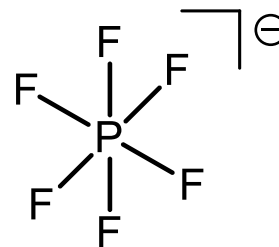
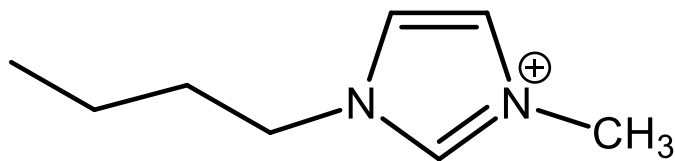
Halide-based (Cl^- , Br^- , F^-)
Halogeno-based (BF_4^- , PF_6^-)
Halogenoaluminium(III)

NO_3^- , ClO_4^- , HSO_4^-
 SO_3^- , CF_3SO_3^-
 $(\text{CF}_3\text{SO}_2)_2\text{N}^-$
 $(\text{CF}_3\text{SO}_2)_3\text{C}^-$

**Organic
cation**

anion

Ionic Liquids - Green Solvents for the Future



$[NR_4]^+$

$[PR_4]^+$

$[SR_3]^+$

cations commonly found in ionic liquids



Table salt $NaCl$ and ionic liquid 1-butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide at 27 °C

The resistance of a liquid to flow is called the _____ of the liquid.

a. density

☒ b. viscosity

c. potential energy

d. flow rate



The energy required to cause a liquid to boil is called the _____ of the liquid.

- a. boiling point
- b. freezing point
- ☒ c. heat of vaporization
- d. heat of fusion



Isopropyl alcohol feels cool to the touch because it has an (X) heat of (Y).

- a. X = exothermic, Y = vaporization
- ☒ b. X = endothermic, Y = vaporization
- c. X = exothermic, Y = fusion
- d. X = endothermic, Y = fusion



When heat is added to ice at zero degrees Celsius, what will happen?

- a. The temperature will increase.
- b. The temperature will decrease.
- ☒ c. The temperature will not change.
- d. A supercritical fluid will form.



The highest temperature at which a substance can exist in its liquid state is called its _____ point.

- a. boiling
- b. freezing
- c. triple
- ☒ d. critical



The temperature and pressure at which all three phases exist simultaneously is called the _____ point of a substance.

- a. boiling
- b. freezing
- ☒ c. triple
- d. critical



At high altitudes, the boiling point of water is

- a. 100 degrees Celsius.
- b. greater than 100 degrees Celsius.
- ☒ c. less than 100 degrees Celsius.
- d. equal to its freezing point.



Substance X has a boiling point of 150 degrees Celsius and a freezing point of 15 degrees Celsius. The condensation point of X

- ☒ a. is 150 degrees Celsius.
- b. is 15 degrees Celsius.
- c. is 165 degrees Celsius.
- d. is 135 degrees Celsius.



Which of the terms below does NOT describe a type of liquid crystal?

- ☒ a. Eutectic
- b. Nematic
- c. Smectic
- d. Cholesteric

